

Bank Bailouts and Moral Hazard: Evidence from Germany

Lammertjan Dam

University of Groningen and CIBIF

Michael Koetter

University of Groningen, CIBIF, and Sveriges Riksbank

We use a structural econometric model to provide empirical evidence that safety nets in the banking industry lead to additional risk taking. To identify the moral hazard effect of bailout expectations on bank risk, we exploit the fact that regional political factors explain bank bailouts but not bank risk. The sample includes all observed capital preservation measures and distressed exits in the German banking industry during 1995–2006. A change of bailout expectations by two standard deviations increases the probability of official distress from 6.6% to 9.4%, which is economically significant. (*JEL* C30, C78, G21, G28, L51)

Deposit insurance schemes protect depositors against losses when a bank fails to meet its debt obligations. Selected banks may also receive capital when in distress, so-called “bailouts,” for instance, when deemed “too-big-to-fail” or because too many banks fail simultaneously (see [O’Hara and Wayne 1990](#); [Acharya and Yorulmazer 2007](#); [Brown and Dinç 2011](#)). Generally, financial safety nets aim to prevent bank runs ([Diamond and Dybvig 1983](#)), reduce the social cost of bank failures ([Gropp et al. 2010](#)), and promote financial stability ([Demirgüç-Kunt and Detragiache 2002](#)). But theoretical studies by [Cordella](#)

We thank an anonymous referee and the editor, Alexander Ljungqvist, for their constructive feedback. We received helpful comments during seminars at Sveriges Riksbank, Deutsche Bundesbank, the European Central Bank, the Universities of Groningen, Leipzig, Limoges, Umeå, Maastricht, Amsterdam (VU), and Essex as well as the Tilburg/EBC conference, the FIRS meetings, the Chicago Fed Conference, the SMYE conference, the International Risk Management Conference in Florence, the Benelux Corporate Finance day, the MFS meetings, the NAKÉ day, and the FMA meetings. Without implicating them, we are indebted to Rob Alessie, Fernando E. Alvarez, Giovanni Dell’Ariccia, Thorsten Beck, Charles Calomiris, Ben Craig, Ferre DeGraeve, Hans Degryse, Bob DeYoung, Valerie de Bruykere, Falko Fecht, Rients Galema, Heinz Herrmann, Tor Jacobsen, Alfred Lehar, Robert Lensink, Aljar Meesters, Loretta Mester, Leonard Nakamura, Steven Ongena, Kasper Roszbach, Klaus Schaeck, Paul Schure, Frank Smets, Veronique Vermoesen, Tom Wansbeek, and Michael Wedow for their feedback. We thank the Bundesbank for providing data. The views expressed here are strictly those of the authors and not of the Bundesbank or Sveriges Riksbank. Teun Brinkman and Karl Harmenberg provided excellent research assistance. This work was supported by the Netherlands Organization for Scientific Research NWO [grant VENI 016.075.164]. All errors are our own. Send correspondence to Michael Koetter, University of Groningen, Faculty of Economics and Business & CIBIF, PO Box 800, 9700 AV Groningen, the Netherlands; telephone: +31 50 318 2665. E-mail: m.koetter@rug.nl. Lammertjan Dam, University of Groningen, Faculty of Economics and Business & CIBIF, PO Box 800, 9700 AV Groningen, the Netherlands. E-mail: L.dam@rug.nl.

© The Author 2012. Published by Oxford University Press on behalf of The Society for Financial Studies. All rights reserved. For Permissions, please e-mail: journals.permissions@oup.com.

doi:10.1093/rfs/hhs056

Advance Access publication April 5, 2012

and Yeyati (2003), Gorton and Huang (2004), and others¹ show that such safety nets also can create moral hazard in the form of excessive risk-taking behavior. We develop a structural model to estimate empirically the extent to which bank bailouts create moral hazard.

The main challenge faced by any empirical test of this moral hazard hypothesis is an identification problem. Bailed-out banks are, by definition, in some sort of distress and exhibit high risk. But this might be simply due to bad luck rather than to bad behavior. Even a prudently managed bank might be distressed because of an exogenous shock. Using the bailout of such an unlucky bank to explain its risk is no evidence for moral hazard. Moral hazard problems arise only from the *additional* risk taking due to higher bailout expectations, which are usually not observable. Previous empirical studies approximate bailout expectations with safety net membership to explain observed bank risk.² Merely regressing bank risk on safety net membership is insufficient to identify moral hazard, because safety net membership alone is neither a guarantee nor a prerequisite to be bailed out. A sufficiently important player might be rescued when in distress irrespective of whether it is a safety net member or not (Freixas and Rochet 2011). Since it is bailout expectations, rather than actual rescues or safety net membership, that matter for risk-taking behavior, we require an estimate of banks' bailout expectations to test the moral hazard hypothesis. Econometrically, the identification of moral hazard therefore requires variables that well explain the likelihood of a bank bailout, but are uncorrelated with the risk taking of banks.

The main contribution made by this article is to use political factors in a structural estimation approach to identify the causal effect of bank bailouts on additional risk taking. We predict bailout expectations conditional on political factors and then test if bailout expectations increase risk taking. Political factors are suited for the identification of moral hazard effects, because they explain bailout probabilities, but do not directly affect risk taking. A number of previous studies analyze the political economy of banking and lending; see, e.g., Sapienza (2004), Khwaja and Mian (2005), Brown and Dinç (2005), and Dinç (2005) for evidence from Italy, Pakistan, and emerging markets, respectively. These studies were criticized because they studied economies with generally weak institutions. Our study complements this

¹ The theoretical literature on the moral hazard effects of deposit insurance, liquidity provision policies, or both is abundant. For example, Repullo (2000) and Boyd et al. (2004) compare deposit insurance and liquidity provision policies. Kareken and Wallace (1978) and Hazlett (1997) look specifically at the moral hazard effects of deposit insurance. More recent studies aim to derive policies to discipline systemically important financial institutions, including limitations to managerial compensation and charging systemic risk taxes (Freixas and Rochet 2011).

² See, for example, Karels and McClatchey (1999), who find no relation between moral hazard and the risk-taking behavior of U.S. credit unions in the 1990s, or Gropp and Vesala (2004), who report that explicit deposit insurance in fact reduced risk for a sample of 128 European banks between 1991–1998. An exception is Keeley (1990), who shows that deposit insurance *payouts* and higher bank risk taking coincided among banks in the United States during the 1980s.

strand of literature because we provide evidence on the importance of political factors in a developed, stable economy to explain the likelihood of bank bailouts.

To this end, we use a novel data set that includes *actual* bailouts rather than mere safety net memberships. The observation of all capital injections in the entire banking industry, as well as (regional) political differences, allows us to identify bailout expectations in the structural model. To measure risk, we rely on the official records of distress as defined by regulatory authorities for all 3,554 German banks during the period 1995–2006. The regulator defines distress as a situation in which “an institution’s existence will be endangered [. . .] without support measures” (Deutsche Bundesbank 2007, p. 75). The probability of distress is thus a more direct measure of risk compared to usually employed accounting-based ratios. Support measures are either exits through restructuring mergers or capital injections. Observations of either bailout or exit thus reveal when the regulator deems the ultimate risk faced by a bank as too high, namely, to fail. On average, 8% of all banks were distressed, and around 6.1% subsequently received a bailout in the form of capital support.

As a second contribution, we test if moral hazard effects due to the existence of safety nets can be mitigated by supervisory policy. We exploit a detailed sample of various supervisory interventions, such as hearings, the dismissal of managers, or penalties, that were applied to both distressed and sound banks. To our knowledge, we are the first to test empirically if such interventions can mitigate moral hazard and to estimate the effectiveness of various supervisory measures.

An increase of bailout expectations by 1% increases risk taking, measured as the likelihood of distress, by 7.1 basis points. A change in bailout expectations from one standard deviation below the mean to one standard deviation above the mean increases predicted distress probabilities from 6.6% to 9.4%. This economically significant increase in risk taking thus provides evidence for moral hazard due to bailout expectations. This result is in line with studies using historical U.S. data for thrifts and banks (Grossman 1992; Wheelock and Wilson 1995) or cross-country samples, including many developing economies (Demirgüç-Kunt and Detragiache 2002). The evidence of moral hazard is robust across a range of alternative measures of risk, out-of-sample prediction approaches, and subsamples. Only supervisory interventions directly targeting bank management and outright restrictions mitigate moral hazard. Weak interventions, such as warnings, do not reduce moral hazard.

We discuss the methodology in Section 1 and present the data in Section 2. Section 3 is devoted to the identification strategy and bailout results. Risk regression results are discussed in Section 4. Section 5 assesses the moral hazard effects of supervisory interventions. We conclude in Section 6.

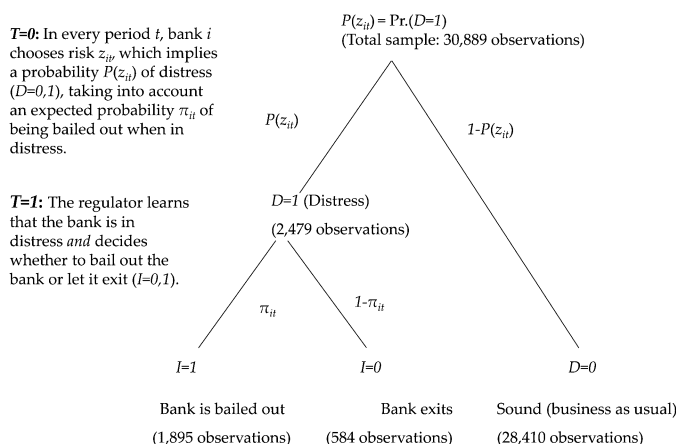


Figure 1
Game tree

1. Structural Model and Econometrics

To identify moral hazard, we cannot just regress banks' risk-taking behavior on observed bailouts because any bailed out bank is distressed. The resulting endogeneity problem requires that we impose some structure on the data to separate bad luck from bad behavior. Like Cordella and Yeyati (2003), we therefore model many simultaneous, static two-player games between a bank and the regulator.³ The timing and choices are depicted in Figure 1.

At $T = 0$, each bank chooses its "riskiness", z_{it} , which maps directly to the probability of "distress" $P(z_{it})$.⁴ The regulator cannot observe this riskiness directly. But at $T = 1$, the consequences of the risk-taking behavior of banks are revealed, and the regulator may declare a bank in distress and weigh the cost of a bailout against the cost of letting a bank exit. *Always* bailing out distressed banks causes severe moral hazard problems (Gale and Vives 2002). But the regulator also cannot credibly "announce" a bailout probability for every bank at $T = 0$ because of a classical commitment problem. For example, if a bank is "too-big-to-fail," it is saved irrespective of any a priori announcement of the regulator.⁵ Only by not revealing which bank "types" it will bail out can it mitigate the moral hazard problem.

Thus, individual banks do not know for sure if they are saved when in distress. But they can infer a bailout probability by relating bailout behavior

³ We describe the authorities referred to as "the regulator" and all events (distress, bailout, and exit) in Section 2.

⁴ Banks do not choose $P(z_{it})$ directly, but business decisions, e.g., on lending or trading, influence it. Different sources of risk may offset one another. In practice, the regulator rates banks on basis of estimated conditional distress probabilities (Deutsche Bundesbank 2007). We assume that each bank chooses z_{it} irrespective of peers.

⁵ A recursive objective and the commitment problem rule out dynamic strategies. Because the game is sequential, a mixed strategy is not optimal. The regulator will play pure strategies, bailout or not.

to observables. To capture the expectations of bank i in year t about the regulator's behavior, we model the expected bailout probability π_{it} conditional on a bank being in distress as

$$\begin{aligned}\pi_{it} &\equiv E[I_{i,t}|D_{i,t} = 1] \\ &= \alpha_{year} + \alpha_{state} + \alpha_{pillar} + X'_{i,t-1}\alpha_1 + M'_{m,t-1}\alpha_2 + Z'_{m,t}\beta.\end{aligned}\quad (1)$$

$I_{i,t}$ is an indicator equal to one if the regulator bails out the bank, and $D_{i,t} = 1$ indicates that the bank is in distress. α denotes fixed effects for years, states, and so-called banking "pillars." $X_{i,t-1}$ and $M_{m,t-1}$ are lagged vectors of observable bank i and macrovariables at the municipality level. $Z_{m,t}$ is a vector of political covariates that explains bailouts, but not risk taking.

At $T = 0$, the bank chooses the risk, z_{it} , that maximizes its expected value, accounting for the expected probability of a bailout in case of distress. The first-order condition is approximated by a probability of distress equation:⁶

$$\begin{aligned}P(z_{i,t}) &= E[D_{i,t}] \\ &= \kappa_{year} + \kappa_{state} + \kappa_{pillar} + \gamma E[I_{i,t}|D_{i,t} = 1] + X'_{i,t-1}\kappa_1 + M'_{m,t-1}\kappa_2.\end{aligned}\quad (2)$$

Again, $D_{i,t}$ indicates if the bank is in distress or is sound. The fixed effects, as well as the variables $X_{i,t-1}$ and $M_{m,t-1}$, are the same as before. The parameter of interest is γ , the coefficient for the expected bailout probability, $E[I_{i,t}|D_{i,t} = 1]$, from Equation (1). We identify, e.g., if large banks take more risk simply because they are large (direct effect through κ_1) or because they are more likely to be bailed out (indirect effect through γ). It is impossible to draw such inferences by regressing probabilities of distress on bank characteristics directly, which constitutes the appeal of our structural model. But we need an exclusion restriction for identification. Therefore, the political variables $Z_{i,t}$ are not specified in Equation (2).

Estimation of this system poses two main challenges. First, the expected bailout probabilities, $E[I_{i,t}|D_{i,t} = 1]$, cannot be observed. Second, because all bailed-out banks are distressed, we face an endogeneity problem.

First, note that the distress Equation (2) corresponds to the top two branches of the game tree in Figure 1, i.e., the full sample. Estimation of this equation requires expected bailout probabilities when in distress for all $N = 30,889$ bank-year observations. But these bailout expectations are *latent*. We can only estimate expected bailout probabilities when these are revealed, i.e., only for banks in distress. Estimation of Equation (1) is thus empirically restricted to a subsample of 2,479 bank-year observations that correspond to the lower left two branches of the game tree in Figure 1. To deal with this restriction and to estimate the structural parameters of the system of simultaneous equations, we apply a two-step procedure. First, we estimate Equation (1)

⁶ One can show that the bank faces a trade-off between maximizing expected cash flows, which depend on the choice of risk and a terminal value. The latter depends also on the probability of distress and bailouts.

to obtain the expected bailout probability conditional on distress for the corresponding subsample of distressed banks. We use the estimates of $\hat{\alpha}$ and $\hat{\beta}$ to calculate the predicted bailout probabilities, $\hat{\pi}_{it}$, for the entire sample of banks, both sound and distressed. Second, we specify this generated regressor to estimate Equation (2). Put differently, we predict out-of-sample bailout expectations from parameters estimated for the subsample of distressed banks to obtain estimates, $\hat{\pi}_{it}$, for the (latent) bailout expectations, $E[I_{i,t}|D_{i,t} = 1]$, for *all* banks. Under standard regularity conditions, the first-step estimation yields consistent estimates of the true parameters. Therefore, we also obtain consistent estimates for the second equation (see [Murphy and Topel 2002](#)).

To solve the second challenge concerning endogeneity and to identify moral hazard effects, we require two assumptions. First, the identifying covariates Z_{it} must be orthogonal to risk taking. Because identification is crucial, we motivate the choices for Z_{it} separately in Section 3. We estimate linear probability models to ensure that identification is based on the exclusion restriction and not on functional form.⁷ Second, it is impossible to observe if a bailed-out bank had survived without support. Therefore, we assume that the regulator only bails out banks that would have exited otherwise.

2. Data and Summary Statistics

2.1 Bailouts and risk

Bailouts are defined as a bank receiving a capital injection from the responsible banking association's insurance fund. Risk taking is measured as the probability of distress. Distress is defined as the event in which the regulator officially declares that the bank's existence will be seriously endangered without support measures ([Deutsche Bundesbank 2007](#)). Support measures are a bailout or the exit of the bank in the form of a restructuring merger. Both events are recorded by the Bundesbank. So, even though risk taking itself is latent, observing these policies reveals those times in which the regulator deems the risk that a bank fails too high.

This risk measure is consistent with U.S. bank hazard studies that estimate the risk of a bank failure conditional on multiple sources of risk rather than only individual drivers, such as credit risk ([Wheelock and Wilson 1995](#)).⁸ Measuring risk as a probability of distress is also in line with the metric used by regulators to rate banks and determine policy.⁹

Table 1 shows the descriptive statistics of distressed and sound banks over time. On average, 8% of all banks were distressed per year between 1995 and

⁷ Throughout, we use two-way clustered standard errors at the bank and state levels ([Petersen 2009](#)).

⁸ Prompt corrective action in the United States implies that troubled banks are immediately closed. In Germany, distressed banks are either supported through recapitalization or merged with healthy banks ([Koetter et al. 2007](#)). This subtle difference between failures and distress reflects preemptive, rather than prompt, corrections.

⁹ See the financial stability reports of Bundesbank since 2004. We also specify alternative risk proxies below.

Table 1
Sound and distressed banks over time

Year	Sound		Distressed				Total
			Bailout		Exit		
	N	% of total	N	% of total	N	% of total	N
1995	3,240	94.2	166	4.8	33	1.0	3,439
1996	3,126	93.9	175	5.3	28	0.8	3,329
1997	3,000	92.6	193	6.0	48	1.5	3,241
1998	2,831	92.1	174	5.7	69	2.2	3,074
1999	2,603	91.5	170	6.0	73	2.6	2,846
2000	2,363	91.1	166	6.4	66	2.5	2,595
2001	2,144	89.6	173	7.2	76	3.2	2,393
2002	1,977	89.5	175	7.9	58	2.6	2,210
2003	1,848	89.6	161	7.8	53	2.6	2,062
2004	1,798	91.5	139	7.1	29	1.5	1,966
2005	1,756	92.5	115	6.1	27	1.4	1,898
2006	1,724	93.9	88	4.8	24	1.3	1,836
Total	28,410	92.0	1,895	6.1	584	1.9	30,889

The table shows frequency distributions for sound and distressed banks with complete cases in the regression analyses. Distress is defined as the occurrence of either a bailout or exit of the bank due to a restructuring merger induced by the regulator. Bailout is defined as a capital injection by the responsible insurance fund of the bank.

2006. Most banks have been bailed out (6.1%), and the share of exits due to restructuring mergers is fairly low (1.9%).

For ease of exposition, we have used so far the term “regulator.” In practice, the decision to bail out or merge a bank involves multiple authorities. First, the German Financial Supervision Authority (BaFin) is responsible for banking supervision in Germany. Second, the central bank is responsible for ongoing supervision. Third, audits for the savings and cooperative banking sectors are conducted by regional banking associations.

Insurance funds reflect the structure of the German banking industry, which features three sectors, or “pillars”: commercial banks, cooperative banks, and savings banks (Krahen and Schmidt 2004). Compulsory and sector-specific, voluntary insurance schemes coexist and are organized per pillar.¹⁰ Insurance protection comprises deposit insurance and institutional warrants. The latter provide the capital injections that we consider.¹¹

Insurance funds generally obtain information on the financial health of member banks from auditors, but prudential supervisors also can directly inform insurance funds if they deem a bank excessively risky. For example, the charter of regional insurance funds in the savings bank sector stipulates that the board of a regional savings association must decide, with a two-thirds majority, whether to declare a member bank distressed.¹² Declaring a member distressed

¹⁰ The Association of German Banks (BdB), the German Savings and Loan Association (DSGV), and the Federal Association of Cooperative Banks (BVR).

¹¹ Note that we do not consider bailouts by the government or the Banking Sector Stabilization Fund (Soffin).

¹² See www.sparkasse.de/s_finanzzgruppe/haftungsverbund/sicherungs_reserve.html.

triggers corresponding support measures, either a restructuring merger or a capital injection. The insurance fund of the association then informs the management and owners of the bank of the decision. They are also required by law to inform the supervisory authorities, BaFin and Bundesbank. Supported banks must reveal any information to the fund to allow for a transparent assessment of their financial situation.

The objectives of the insurance schemes and official supervisors may not always be in line. But eventually, all institutions involved collectively contribute to the decision of whether or not to bail out a bank. For our purposes, we do not need to know *how* the bailout decision is made, but only *whether* it is made. We simply test if the actions of safety nets induce moral hazard.

2.2 Covariates

We use a statistical selection procedure to select bank-specific (X_{it-1}) and macro- (M_{mt-1}) covariates with high predictive power of bailouts and distress. All covariates are lagged by one year to mitigate simultaneity concerns. Table 2 provides definitions of all variables. The first two panels of Table 3 show the

Table 2
Variable definitions and sources

	Definition	Level	Unit	Source
Bank-specific covariates (X)				
Total assets $_{t-1}$	Gross total assets (GTA)	bank	billion €	BBK
Hidden reserves $_{t-1}$	Hidden reserves due to below market value valuations, according to §340f commercial code (HGB), as share of GTA	bank	%	BBK
Nonperforming loan share $_{t-1}$	Latent risk loans relative to the total of audited loans	bank	%	BBK
Customer loan share $_{t-1}$	Lending volume to private and nonfinancial corporate customers relative to total earning assets	bank	%	BBK
Return on equity $_{t-1}$	Operating return relative to gross total equity	bank	%	BBK
Cost-income ratio $_{t-1}$	Total administrative expenditures to total operating revenue	bank	%	BBK
Liquid asset share $_{t-1}$	Cash and overnight interbank assets relative to GTA	bank	%	BBK
Regional market share $_{t-1}$	GTA per bank relative to aggregate GTA in municipality ("Kreis")	bank	%	BBK
Regional macro covariates per municipality ("Kreis", M)				
Corporate insolvencies $_{t-1}$	Corporations that filed for bankruptcy relative to all corporations	municipality	%	BBR, SSO
Unemployment rate $_{t-1}$	Registered unemployed workers and employees relative to the social-security insured population	municipality	%	BBR, SSO
Real output per capita growth $_{t-1}$	Growth of gross municipality output per capita in 2005 prices	municipality	%	BBR, SSO

(continued)

Table 2
Continued

	Definition	Level	Unit	Source
Identifying covariates (Z)				
Election	State parliament election	state	dummy	SSO, EC
Vote share difference	The difference between the vote share of total votes cast for the governing coalition less the vote share cast for the strongest opposition party (see Dinç and Gupta 2011) per municipality ("Kreis") in state parliament elections.	municipality	%	SSO, EC
Same party	State and federal prime ministers are from the same party	state	dummy	SSO, FSO, EC
Alternative risk measures and components				
Nonperforming loan share	Latent risk loans relative to the total of audited loans	bank	%	BBK
Equity capital ratio (CAR)	Equity capital relative to risk-weighted assets (RWA)	bank	%	BBK
Return on risk weighted assets (RoA)	Net operating income relative to RWA	bank	%	BBK
σ_{RoA}	Standard deviation of RoA over preceding three years	bank	%	BBK
Z-score	Following Laeven and Levine (2009) , $(RoA + CAR)/\sigma_{RoA}$	bank	points	BBK
Net fixed interest rate assets (NFIRA)	Fixed interest-bearing assets less fixed interest-bearing liabilities	bank	billion €	BBK
Fixed interest rate gap (FIRG)	Average interest rate on fixed rate assets less fixed rate liabilities	bank	%	BBK

BBK denotes the German central bank ("Bundesbank"). BBR denotes the Federal Office for Construction and Urban Planning ("Bundesamt für Bauwesen und Raumordnung"). SSO denotes the 16 State Statistical Offices ("Statistische Landesämter"). FSO denotes the Federal Statistical Office. EC denotes election council per state. Elections are held every four to five years, but at different dates in each of the sixteen federal states. See Table A for details on the distribution across states and time. There are 457 municipalities in the sample.

means and standard deviations of the bank-specific covariates and regional macro-covariates per type of event.

The subsamples are not too heterogeneous, which is comforting because we use predicted bailout probabilities from Equation (1) to estimate the moral hazard effect on risk taking in Equation (2).

The importance of a bank for the financial system is a key determinant for both bailouts and risk taking. Negative externalities can be substantial if a large and highly connected bank fails. We control for both gross total assets and the market shares per municipality (Kreis). Hidden reserves are used to smooth income and are an important indicator of solvency risk. They result from mark-below-market valuation, in line with §340f of the German commercial code (HGB), and are measured as a share of total assets. Nonperforming loans measure credit risk and are defined as the share of all audited loans that are at latent risk relative to total audited loans. The customer loan share is household and corporate loans relative to total earning assets. Profitability is measured as operating return relative to total equity. The cost-income ratio relates general

Table 3
Descriptive statistics, explanatory, and identifying covariates

Variable	Sound			Bailout			Exit		
	Mean	SD	N	Mean	SD	N	Mean	SD	N
Bank-specific covariates (<i>X</i>)									
Total assets _{<i>t</i>-1}	0.67	1.26	28,410	0.57	1.03	1,895	0.39	0.94	584
Hidden reserves _{<i>t</i>-1}	0.35	0.57	28,410	0.08	0.22	1,895	0.12	0.35	584
Nonperforming loan share _{<i>t</i>-1}	9.82	8.82	28,410	13.02	9.85	1,895	11.55	10.36	584
Customer loan share _{<i>t</i>-1}	62.30	14.65	28,410	61.95	15.56	1,895	61.64	17.19	584
Return on equity _{<i>t</i>-1}	22.25	12.51	28,410	16.00	14.67	1,895	13.97	15.46	584
Cost-income ratio _{<i>t</i>-1}	68.97	10.41	28,410	67.79	16.09	1,895	71.31	48.01	584
Liquid asset share _{<i>t</i>-1}	2.07	1.16	28,410	2.41	1.15	1,895	2.20	1.14	584
Regional market share _{<i>t</i>-1}	15.26	21.96	28,410	15.42	20.84	1,895	9.59	17.26	584
Regional macro covariates per municipality ("Kreis", <i>M</i>)									
Corporate insolvency rate _{<i>t</i>-1}	0.89	0.39	28,410	1.08	0.54	1,895	0.97	0.47	584
Unemployment rate _{<i>t</i>-1}	9.53	3.90	28,410	11.27	5.24	1,895	10.31	4.76	584
Real output per capita growth _{<i>t</i>-1}	2.63	5.06	28,410	3.63	7.52	1,895	3.21	8.29	584
Identifying covariates (<i>Z</i>)									
Vote share difference	21.62	17.74	28,410	25.68	16.27	1,895	23.46	15.51	584
Election	0.22	0.41	28,410	0.20	0.40	1,895	0.25	0.43	584
Same party	0.50	0.50	28,410	0.44	0.50	1,895	0.43	0.50	584
Alternative risk measures and components									
Nonperforming loan share	9.40	8.48	28,410	12.08	9.64	1,895	10.57	9.13	557
Return on risk weighted assets (RoA)	0.99	1.07	28,409	0.55	1.18	1,893	0.49	1.62	545
σ_{RoA}	0.41	0.70	28,410	0.57	0.75	1,895	0.52	0.88	557
Equity capital ratio	5.54	3.35	28,405	5.35	4.82	1,893	5.49	4.88	541
Z-score	21.68	11.30	28,404	15.11	9.29	1,893	19.74	15.76	541
Net fixed interest rate assets (NFIRA)	22.94	12.08	25,864	18.29	12.91	1,768	17.91	13.40	524
Fixed interest rate gap (FIRG)	1.33	0.54	25,785	1.22	0.55	1,762	1.20	0.61	521

Variables definitions, level of analysis, measurement units, and sources are as shown in Table 2.

administrative expenditures to operating revenues and measures managerial skill. To control for liquidity risk, we specify the sum of cash and overnight interbank assets to total assets.

We control for the health of the corporate sector, the economic cycle, and the financial stance of households using three macrovariables at the municipality level: the share of enterprises that filed for bankruptcy relative to all firms, annual growth of real output per capita, and unemployment rates. In addition, we include state, year, and banking-pillar dummies.¹³

3. Identification

3.1 Identifying covariates

Valid exclusion restrictions, i.e., one or more covariates that explain bailouts well but are uncorrelated with risk-taking behavior, are crucial for the identification of moral hazard in the system of equations. The choice of three

¹³ There are sixteen states (Bundesländer) in Germany and 457 municipalities (Kreise).

identifying covariates is inspired by studies emphasizing the importance of politicians in economic decisions, such as lending, bailouts, and corporate privatization (Dinç 2005; Brown and Dinç 2005; Dinç and Gupta 2011).¹⁴

Political considerations are likely to be important in the German banking industry for three reasons. First, board members of both the BaFin and the Bundesbank are suggested by politicians. Formally, the Ministry of Finance appoints board members. But, e.g., Bundesbank board members are suggested by (alternating) state governments. Second, around 20% of all banks are (regional) government-owned savings banks. Local politicians serve on the supervisory boards of savings banks and influence regional associations, which both audit member banks and maintain regional insurance funds. Third, (local) politicians often serve on the supervisory boards of commercial and cooperative banks, too. We use three political indicators to approximate political influence on bailout decisions.¹⁵

The first relates to an arguably important event in influencing the behavior of politicians: elections. Governments that hold a stake in the banking system may abuse their influence to pursue non-value-maximizing objectives to realize political objectives, such as re-election (Sapienza 2004; Brown and Dinç 2005). We create a dummy variable equal to one if there was a state parliament election in a given year. Dinç and Gupta (2011) similarly show that the privatization of Indian enterprises is more likely if political competition from the main opposition is low. They measure political competition by the margin of votes earned by the governing coalition in a region compared with the vote share of the main opposition coalition. The second political variable is therefore the vote share difference in *state* parliament elections using votes cast at the *municipality* level. The third political variable measures political similarity. We specify an indicator variable that is equal to one if the federal and state prime minister are from the same party. If this is the case, party politics, rather than economic considerations in the region of the bank, can influence bailout decisions.¹⁶

The third panel, “Exclusion restrictions,” in Table 3 shows the mean and standard deviations for the three political variables per event group. The subsamples are not too heterogeneous, which mitigates concerns on out-of-sample prediction. More importantly, elections, vote share differences, and shared party membership vary both over time and across regions. This variation reflects that elections are not held in each state at the same time. On average,

¹⁴ Further tests for a range of alternative political indicators are available upon request.

¹⁵ Preferable direct information on the political membership of bank managers and directors is unavailable.

¹⁶ The 2001 Berliner Bankgesellschaft scandal provides anecdotal evidence. A conservative (CDU) member of Berlin’s state parliament was also a manager at the bank and systematically covered up its poor financial state. Berlin’s prime minister at the time, Eberhard Diepgen, was a conservative, whereas the federal government was headed by Social Democrat Gerhard Schröder (SPD), who led a federal coalition with the Green Party. As rigged practices at the bank leaked to the public, the socialistic and green opposition in the state of Berlin, supported by federal party peers, started a no-confidence motion that brought down the Diepgen administration.

state parliament elections are held every four to five years.¹⁷ Table A in the Appendix shows a detailed breakdown of the spatial and temporal distribution of the political variables. This variation over both time and in the cross-section renders it unlikely that these variables merely reflect a regional or a macroeconomic effect, for which we control explicitly.

3.2 First-stage results

Table 4 shows estimation results for the linear probability model with two-way clustering (bank and state; see Petersen 2009) for the bailout Equation (1). Centered R^2 are around 10% (uncentered R^2 around 79%) for all estimations, which confirms that bailouts are fairly predictable.

The first six columns in Table 4 show that all three political variables are significant, both individually and in various combinations. We control for state, year, and banking group fixed effects, as well as macrovariables at the municipality level. Therefore, these political variables contain additional information on bailout expectations. In all specifications, the indicator for state parliament elections is significantly negative. Bank bailouts seem unpopular among political constituents, and they are significantly less likely during election periods. Larger vote share differences further amplify the reduction of bailout expectations, potentially indicating that more powerful state governments are better able to withstand bailout interests of the opposition and/or other stakeholders. Shared party membership of state and federal leaders is significantly negative as well. A change in vote share differences by one standard deviation paired with an election and shared party membership reduces the expected bailout probability by 13%, which is substantial, given a mean of 76% ($= 585 / (585 + 1,895)$; see Table 1).

Most bank and macro-covariates exhibit plausible signs. Larger banks with high local market shares are more likely to receive a bailout, which corroborates the well-documented too-big-to-fail effect (O'Hara and Wayne 1990). Higher insolvency risk, as reflected by lower hidden reserves, and more credit risk, as measured by the nonperforming loan shares, are also associated with a higher expected bailout probability. Profitability, customer loans, and liquidity are not significant. Regarding macroeconomic factors, only a fragile corporate sector, measured by corporate insolvencies, increases the expected bailout probabilities of banks. Repullo and Martinez-Miera (2010) show that risk (and competition) in banking interacts with corporate sector stability, and regulators may aim to avoid aggravating corporate distress by letting banks fail if the nonfinancial sector is weak.

More important than possible narratives on the reasons for the direction of political indicators is the result that these variables significantly explain

¹⁷ Some states have lengthened the electoral cycle since 1995. There are only two "surprise elections" (Berlin 2001 and Hamburg 2004), concerning thirteen bank-year observations in the distressed sample that do not affect results.

Table 4
First-stage bailout regressions: Identification of bailout probabilities

Variables	Votes	Elections	Same party	Votes and elections	Votes and same party	All three instruments	Election lags and leads	Spatial weights	Bruxelles concordance
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Vote share difference _{<i>t</i>}	−0.002** [0.001]			−0.002** [0.001]	−0.002** [0.001]	−0.002** [0.001]	−0.001** [0.001]		−0.002** [0.001]
Election _{<i>t</i>}		−0.042** [0.019]		−0.039** [0.019]		−0.047*** [0.017]	−0.062** [0.028]		−0.047*** [0.017]
Same party _{<i>t</i>}			−0.041* [0.023]		−0.039* [0.021]	−0.046** [0.020]	−0.047** [0.019]	−0.047** [0.020]	−0.046** [0.020]
Election _{<i>t</i>−1}							−0.044 [0.035]		
Election _{<i>t</i>−2}							−0.014 [0.033]		
Election _{<i>t</i>+1}							−0.002 [0.036]		
Election _{<i>t</i>+2}							−0.003 [0.032]		
Spatial weight elections _{<i>t</i>}								−0.067*** [0.021]	
Spatial weight vote share differences _{<i>t</i>}								−0.001* [0.001]	
Preconcordance (2001) indicator _{<i>t</i>}									−0.055* [0.031]
Total assets _{<i>t</i>−1}	0.066*** [0.022]	0.068*** [0.023]	0.067*** [0.023]	0.067*** [0.023]	0.065*** [0.022]	0.066*** [0.023]	0.065*** [0.022]	0.067*** [0.023]	0.066*** [0.023]
Hidden reserves _{<i>t</i>−1}	−0.106*** [0.027]	−0.105*** [0.027]	−0.106*** [0.028]	−0.105*** [0.027]	−0.106*** [0.028]	−0.105*** [0.027]	−0.101*** [0.028]	−0.105*** [0.028]	−0.105*** [0.027]
Nonperforming loan share _{<i>t</i>−1}	0.003*** [0.001]	0.003*** [0.001]	0.003*** [0.001]	0.003*** [0.001]	0.003*** [0.001]	0.003*** [0.001]	0.003*** [0.001]	0.003*** [0.001]	0.003*** [0.001]
Customer loan share _{<i>t</i>−1}	0.001 [0.001]	0.001 [0.001]	0.001 [0.001]	0.001 [0.001]	0.001 [0.001]	0.001 [0.001]	0.001 [0.001]	0.001 [0.001]	0.001 [0.001]

(continued)

Table 4
Continued

Variables	Votes	Elections	Same party	Votes and elections	Votes and same party	All three instruments	Election lags and leads	Spatial weights	Bruxelles concordance
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Return on equity _{<i>t</i>−1}	0.001 [0.001]	0.001 [0.001]	0.001 [0.001]	0.001 [0.001]	0.001 [0.001]	0.001 [0.001]	0.001 [0.001]	0.001 [0.001]	0.001 [0.001]
Cost-income ratio _{<i>t</i>−1}	−0.001*** [0.000]	−0.001*** [0.000]	−0.001*** [0.000]	−0.001*** [0.000]	−0.001*** [0.000]	−0.001*** [0.000]	−0.001*** [0.000]	−0.001*** [0.000]	−0.001*** [0.000]
Liquid asset share _{<i>t</i>−1}	0.014 [0.010]	0.014 [0.010]	0.014 [0.009]	0.013 [0.010]	0.013 [0.009]	0.013 [0.009]	0.013 [0.009]	0.013 [0.009]	0.013 [0.009]
Regional market share _{<i>t</i>−1}	0.003*** [0.001]	0.003*** [0.001]	0.003*** [0.001]	0.003*** [0.001]	0.003*** [0.001]	0.003*** [0.001]	0.003*** [0.001]	0.003*** [0.001]	0.003*** [0.001]
Corporate insolvency rate _{<i>t</i>−1}	0.124*** [0.041]	0.102*** [0.038]	0.107*** [0.034]	0.127*** [0.043]	0.133*** [0.039]	0.137*** [0.040]	0.136*** [0.038]	0.125*** [0.036]	0.137*** [0.040]
Unemployment rate _{<i>t</i>−1}	0.003 [0.003]	0.004 [0.003]	0.004 [0.003]	0.003 [0.003]	0.004 [0.004]	0.004 [0.004]	0.004 [0.004]	0.004 [0.003]	0.004 [0.004]
Real output per capita growth _{<i>t</i>−1}	0.000 [0.002]	−0.001 [0.002]	0.000 [0.002]	0.000 [0.002]	0.000 [0.002]	0.000 [0.002]	0.000 [0.002]	0.000 [0.002]	0.000 [0.002]
Constant	0.537*** [0.103]	0.513*** [0.107]	0.497*** [0.104]	0.534*** [0.107]	0.518*** [0.106]	0.513*** [0.112]	0.525*** [0.126]	0.514*** [0.114]	0.513*** [0.112]
Centered <i>R</i> ²	0.105	0.104	0.104	0.106	0.106	0.108	0.109	0.107	0.108
Uncentered <i>R</i> ²	0.789	0.789	0.789	0.789	0.789	0.79	0.79	0.79	0.79
Pseudo-Kleibergen-Paap F-statistic	5.97	4.92	3.28	9.143	12.64	15.03	45.04	16.73	36.29
Pseudo-Hansen J-statistic	n/a	n/a	n/a	3.269	0.029	4.283	4.588	4.906	4.283
<i>p</i> -value of Hansen	n/a	n/a	n/a	0.071	0.864	0.118	0.101	0.179	0.118

Linear probability model with an indicator variable equal to one if a bank was bailed out and zero if it exited. Between 1995 and 2006, there are 2,479 observations for distress (bailout or exit). Explanatory variables are defined in Table 2. Vote share differences are the share of votes obtained by the majority party/coalition per municipality during state parliament elections less the vote share of the strongest opposition party in the municipality (Ding and Gupta 2011). Elections are an indicator variable equal to one if state parliament elections were held in the municipality in which the bank is headquartered during that year. Same party is an indicator variable equal to one if the party of the state and federal prime minister in which a bank is headquartered is the same. Spatial weights denotes variables, Elections and Vote share differences, that are weighted with an exponential distance-decay function based on geographical distances between geographical centers of each of the 457 municipalities in Germany. Preconcordance is an indicator variable equal to one in all periods prior to the Bruxelles Concordance of 2001, which abolished the explicit guarantees for German savings banks that are owned by regional governments. The F-statistic is the Kleibergen-Paap test for weak identification pertaining to the excluded covariates in the respective Columns (Baum et al. 2007). The Hansen J is a test statistic for overidentification (Baum et al. 2007). Two-way clustered standard errors at the state and bank level are in brackets (Petersen 2009). Fixed effects are for states (“Bundesländer”), years, and banking sectors (commercial, savings, and cooperative). *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively. n/a denotes not applicable.

bailouts, whereas it is unlikely that they substantially, directly affect banks' risk taking. Political variables thus aid in the identification of the effect of interest: moral hazard. But individually significant coefficients are insufficient in validating the exclusion restrictions. The econometric approach, a nested system of two equations, exhibits many similarities to classic instrumental variable (IV) estimation, though the identifying covariates are technically not "instruments." Their purpose is to generate the latent bailout expectations, which we cannot observe/estimate for banks that are not in distress, as opposed to mitigating a direct endogeneity problem for observed variables in a classical IV framework. However, to be validly excluded, our identifying covariates require similar properties to that of instruments, namely 1) to possess sufficient explanatory power for the endogenous variable, i.e., bailouts; and 2) to be orthogonal to the dependent variable in the second equation, i.e., risk taking. Therefore, we adopt two econometric tests regarding explanatory power to explain bailouts and orthogonality with the dependent variable, as well as logical validation tests of the political variables.

3.3 Weak identification

First, we are concerned with weak identification, i.e., whether political variables explain *sufficient* variation in bailouts to identify the effect of bailout expectations on risk taking (Stock et al. 2002; Baum et al. 2007). Significance of the political variables alone is not the best indication of whether they possess sufficient explanatory power. To assess whether these variables jointly generate enough variation in bailout expectations, we perform a pseudo-Kleibergen-Paap F-test. This is in fact the standard Kleibergen-Paap F-test, were it not that the sample size for our second-stage regressions is larger. Though the Kleibergen-Paap F-test technically does not rely on the second-stage regressions, we give it the label "pseudo" to remind the reader that this test can only validate how well the political variables explain bailout expectations for the subsample of banks that are in distress. It is impossible to know if this result also holds for the full sample. However, we can validate the political variables to the extent that rejection of sufficient explanatory power for the subsample implies insufficient power for the whole sample as well, because predicted values of $\hat{\pi}$ would have too little variation, which is necessary for proper identification.

Whereas the specifications of individual identifying covariates in Columns (1) to (3) clearly fall short of the rule-of-thumb critical value of ten (Stock et al. 2002), the joint specification of political variables yields F-statistics that corroborate their relevance in explaining bailouts. The F-statistic for the specification in Column (6) with all three political covariates is 15.03 and thus strongly supports this choice.¹⁸ Henceforth, we consider this specification.

¹⁸ For three instruments, the critical value for a bias of less than 10% relative to ordinary least squares at the 5% level is 9.08 (Stock et al. 2002, Table 1). The critical F-statistic for the actual size of a 5% test below 15%, the

3.4 Orthogonality of identifying covariates

The second concern relates to the exogeneity of identifying covariates, i.e., whether the political variables are uncorrelated to (direct) risk taking. Contrary to the Kleibergen-Paap F-test, overidentification tests commonly rely on second-stage estimations. In Table 4, we only report those test statistics that validate the orthogonality of the political instruments and discuss corresponding results in more detail in Section 4.

Specifically, we adopt the Hansen test with the joint null hypothesis that identifying covariates are uncorrelated with the error term and correctly excluded from the second-stage estimation. The difference between what we label pseudo-Hansen J-statistics and the standard test is that the error term from the second-stage regression is calculated slightly differently. In an IV framework, the overidentifying restrictions test pertains to the correlation between the error term of the second-stage regression and the instruments. The error term is calculated using the observed values of the endogenous variable and not the fitted values of the endogenous variable. In our case, we do not observe the bailout expectations for the entire sample. In fact, the whole purpose of our framework is to identify these.

Hence, to perform the analogue of the Hansen J-test, we substitute the *fitted* values of the endogenous variable for the actual values of the endogenous variable. This implies that there generally will be more variation in the generated residuals, rendering this pseudo-Hansen J-test somewhat weaker. Given this concern, we assess the power of the pseudo-statistic with 8,000 simulations, for various specifications and sample sizes. These simulations reveal that for sample sizes above 10,000 and subsamples of the first stage of about 5%–10% of total sample size, the pseudo-test does not underperform, in terms of drawing correct inference, compared to the regular Hansen J-test when the full sample is also available for the first stage.¹⁹ In fact, for sample sizes of 10,000 and above, the pseudo-test is very close to 100% correct inference. Table 4 shows that the pseudo-Hansen J-test is insignificant for all specifications. We can therefore not reject that our political variables are exogenous to risk taking.

3.5 Robustness checks of political variables

Despite econometric testing, the identifying covariates may still fail to measure the relation that we are interested in between politics and bank bailouts. Therefore, we perform three additional robustness checks.

The first test is inspired by Cornelli et al. (2011). If elections influence the bailout decision because politicians behave differently, more distant future

weak-instruments thresholds, is 12.83. Some additionally tested political covariates are individually significant, but did not add information once the three presented political indicators are included. Results are available upon request.

¹⁹ Tables of simulation results and Stata code to generate these simulations are available upon request.

and past elections should affect current bailout expectations less, if at all. Therefore, we simultaneously specify leads and lags of the election variable. Column (6) in Table 4 confirms that all leads and lags of elections are insignificant. The three contemporaneous effects remain intact.

Second, the baseline specification explains bailout expectations with municipality and state-specific political indicators. Political factors in the banks' region should matter the most, but neighboring regions' politics are likely to have an impact as well (see also Huang 2008). For example, even elections in distant regions may change local bailout expectations if politicians interact within parties and across geographical borders. Also, political regions do not coincide with regulatory jurisdictions or the responsible insurance scheme regions.²⁰ Thus, political variables at the municipality (vote share difference) and state level (elections and shared party) pertain to a different spatial "grid" than do bailout choices, which may add "spatial" noise when estimating the parameters of identifying covariates.

Therefore, we specify so-called spatial lags of the election dummy and the vote share difference (Anselin 1988). We multiply the election and vote share difference variables with a weighting matrix based on geographical distances using a standard distance-decay function.²¹ Spatially lagged identifying covariates for each bank receive the full weight, and more distant municipalities receive increasingly less weight.

Spatially weighted elections and vote share differences are also significant and exhibit the same signs and very similar magnitudes as before. Coefficients for weighted and unweighted political variables are not significantly different from each other. Hence, politics in neighboring regions are statistically significant but bear qualitatively identical implications. The explanatory power of these proxies regarding bailout decisions is not driven by spurious correlation due to a lack of geographical congruence of regions. The pseudo-Kleibergen-Paap F-test and Hansen J-statistic confirm the validity of the identifying covariates.

A third concern is that the identifying political variables may actually measure legal changes that are often used in the finance and banking literature as exogenous frictions to identify causal relationships (see, e.g., Cornelli et al. 2011; Strahan and Rice 2010). The German Banking Act (Kreditwesengesetz, KWG) applies federally, and there are no state-specific laws. But an important legal change is the 2001 ruling of the European Court in response to a lawsuit filed by the European Commission against the Federal Republic of Germany in April 2000. The commission argued that explicit regional and

²⁰ There are nine regional Bundesbank branches conducting ongoing supervision, but sixteen states. Cooperative and commercial insurance schemes are organized nationally, whereas there exist eleven regional schemes for savings banks.

²¹ The distance matrix between banks is based on the latitude and longitude of the geographical center of each municipality where the bank is headquartered. We follow the regional science literature and use $\exp(-k * \text{distance})$ with $k = 0.04$ based on simulations.

federal government guarantees of government-owned savings banks distorted competition and violated the EU Treaty. The EU court agreed and ruled in its “Bruxelles Concordance” of 2001 to abolish government guarantees after a grandfathering period by July 2005.

Government guarantees of banks have important implications also for competing, noninsured banks (Gropp et al. 2010). If bailout decisions were in fact driven by this legal change and political indicators merely pick up an omitted variable, we expect their significance to explain bailouts to vanish. We show in Column (9) that all identifying covariates remain significant when including simultaneously an indicator variable equal to one for all periods prior to the 2001 Bruxelles Concordance, which itself is significant at the 10% level. Thus, the three political identifying covariates explain a significant portion of bailouts independently.²²

4. The Effect of Bailout Expectations on Risk Taking

4.1 Moral hazard, expectation formation horizons, and out-of-sample prediction

Table 5 shows the estimation results of the distress Equation (2). Standard errors are two-way clustered at the bank and state levels (Petersen 2009), and we include state, year, and banking pillar fixed effects.

The result in Column (1) is based on the entire sample from 1995 until 2006. The fit of the linear probability model for the distress equation is good. Centered R^2 is around 10%, and uncentered R^2 is around 19%.²³

The coefficient of interest for $\hat{\pi}$ is significantly positive, which confirms the existence of moral hazard due to safety nets. An increase of expected bailout probabilities by 1% increases the likelihood of distress by 0.22%. To put this magnitude into perspective, the mean of the expected bailout probability is 69% with a standard deviation of approximately 17% (see Table 5). Thus, an increase in bailout expectations from one standard deviation below the mean to one standard deviation above the mean implies an increase in risk of 7.5% ($= 0.22 \times 2 \times 17\%$), which is substantial, given an average distress probability of 8%.

Signs and magnitudes of other risk drivers are in line with both the prior evidence of bank distress in Germany (Koetter et al. 2007) and expectations. Larger banks with more hidden reserves, higher profitability, and regional market shares are less risky. Banks with more nonperforming and customer loans, higher managerial inefficiency in terms of cost-income ratios, and more low-yield liquidity are, in turn, more risky.

²² Below we use rolling windows that reflect electoral cycles to predict bailouts, which precludes the specification of the indicator in the exclusion restriction.

²³ Alternative probit model estimations yield very similar pseudo- R^2 . Discriminatory power, reflected by the values of the area under the receiver operating characteristics curve of around 75%, is high. Results are available upon request.

Table 5
Second-stage risk regression: Out-of-sample prediction of bailout expectations

Specification	Pooled		Rolling window (1999-2006)			Out-of-sample prediction		
	(1)	(2)	(3)	(4)	(5)	Common support		Exclude
	1995–2006	1999–2006	5 years	4 years	Recursive	RW	RW & BBK	$\hat{\pi} \in [0,1]$
Bailout expectations $\hat{\pi}$	0.218*** [0.054]	0.19*** [0.071]	0.062*** [0.017]	0.035** [0.014]	0.153*** [0.048]	0.071*** [0.022]	0.052*** [0.018]	0.108*** [0.021]
Total assets _{<i>t</i>−1}	−0.011*** [0.004]	−0.01** [0.005]	−0.003 [0.003]	−0.001 [0.004]	−0.008* [0.004]	−0.004 [0.003]	0.002 [0.006]	0.001 [0.006]
Hidden reserves _{<i>t</i>−1}	−0.024** [0.010]	−0.042*** [0.010]	−0.051*** [0.006]	−0.056*** [0.010]	−0.039*** [0.007]	−0.064*** [0.009]	−0.091*** [0.011]	−0.052*** [0.011]
Nonperforming loan share _{<i>t</i>−1}	0.001*** [0.000]	0.001*** [0.000]	0.002*** [0.000]	0.002*** [0.000]	0.001*** [0.000]	0.002*** [0.000]	0.002*** [0.001]	0.002*** [0.001]
Customer loan share _{<i>t</i>−1}	0.001*** [0.000]	0.001*** [0.000]	0.001*** [0.000]	0.001*** [0.000]	0.001*** [0.000]	0.001*** [0.000]	0.001*** [0.000]	0.001*** [0.000]
Return on equity _{<i>t</i>−1}	−0.003*** [0.000]	−0.005*** [0.000]	−0.004*** [0.000]	−0.004*** [0.001]	−0.005*** [0.000]	−0.004*** [0.000]	−0.005*** [0.001]	−0.004*** [0.001]
Cost-income ratio _{<i>t</i>−1}	0.001*** [0.000]	0.001*** [0.000]	0.001*** [0.000]	0.001* [0.000]	0.001*** [0.000]	0.001*** [0.000]	0.001** [0.000]	0.001** [0.000]
Liquid asset share _{<i>t</i>−1}	0.006** [0.003]	0.005 [0.003]	0.006* [0.003]	0.006* [0.003]	0.006 [0.003]	0.006* [0.004]	0.006* [0.003]	0.005*** [0.003]
Regional market share _{<i>t</i>−1}	−0.001*** [0.000]	0.001* [0.000]	0.000 [0.000]	0.000 [0.000]	0.000 [0.000]	0.000 [0.000]	0.000 [0.000]	0.000 [0.000]
Corporate insolvency rate _{<i>t</i>−1}	0.000 [0.014]	0.018 [0.015]	0.043*** [0.008]	0.046*** [0.016]	0.027*** [0.010]	0.05*** [0.010]	0.058*** [0.018]	0.045*** [0.014]
Unemployment rate _{<i>t</i>−1}	0.004** [0.002]	0.004** [0.002]	0.005** [0.002]	0.005** [0.002]	0.005** [0.002]	0.005** [0.002]	0.006*** [0.002]	0.005** [0.002]
Real output per capita growth _{<i>t</i>−1}	0.000 [0.000]	−0.001 [0.001]	−0.001** [0.001]	−0.001 [0.001]	−0.001 [0.001]	−0.001* [0.001]	−0.001 [0.001]	−0.001** [0.001]
Constant	0.001 [0.061]	0.066 [0.073]	0.107 [0.075]	0.121 [0.080]	0.071 [0.073]	0.097 [0.076]	0.125 [0.086]	0.047 [0.080]

(continued)

Table 5
Continued

Specification	Pooled		Rolling window (1999–2006)			Out-of-sample prediction		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	1995–2006	1999–2006	5 years	4 years	Recursive	Common support		Exclude
						RW	RW & BBK	$\hat{\pi} \in [0,1]$
Descriptive statistics of predicted bailouts $\hat{\pi}$ and regression diagnostics								
Observations	30,889	17,806	17,806	17,806	17,806	17,535	14,288	17,077
Mean of $\hat{\pi}$	0.685	0.661	0.646	0.652	0.644	0.653	0.642	0.635
SD of $\hat{\pi}$	0.169	0.172	0.211	0.221	0.19	0.202	0.327	0.179
Centered R^2	0.098	0.105	0.105	0.105	0.105	0.107	0.11	0.105
(Average) Pseudo-F-statistic	15.03	15.03	11.88	12.04	31.19	11.88	60.04	11.88
Pseudo-Hansen J-statistic	4.283	1.689	1.633	1.435	1.399	2.068	1.9	1.911
p-value of Hansen	0.118	0.430	0.442	0.488	0.497	0.356	0.387	0.464

Linear probability model with an indicator variable equal to one if a bank was distressed. Explanatory variables are defined in Table 2. Columns labeled “Pooled” show distress equation estimations based on bailout regressions on the full sample pertaining to the sample periods indicated in the line-labeled specification. Columns labeled “Rolling window” are based on first-stage regressions for four and five years per estimation window and one recursively increasing sample (see Table A2 and the Online Appendix for first-stage estimation results). The columns labeled “Out-of-sample prediction” are based on samples that restrict the prediction of first-stage estimates of bailout expectations only to those sound banks that are in the common support of propensity score matching conditional on all explanatory covariates per rolling window (RW), rolling window *and* Bundesbank jurisdiction (RW & BBK), and those sound banks with predicted bailout probabilities on the interval between zero and one. For specifications (3) through (8), the first-stage F-statistic is averaged over the respective sample windows used in the bailout equations. Hence, it is identical for specifications (3), (6), and (8). Two-way clustered standard errors at the state and bank level are in brackets (Petersen 2009). Fixed effects are for states (“Bundesländer”), years, and banking sectors (commercial, savings, and cooperative). *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

The remainder of Table 5 tackles two important concerns. First, the multiple institutions that we coin “the regulator” may behave strategically and change their behavior over time. For instance, they may turn hard-nosed, even on large bank bailouts, after a wave of rescue measures in the past. Therefore, the weights on covariates estimated in the bailout regression may change and should not depend on any future information when estimated. We therefore pursue rolling-window approaches. The second concern relates to the different sample sizes available to estimate Equations (1) and (2). The sample of sound banks, for which we infer bailout expectations based on estimated parameters pertaining to the subsample of distressed banks, may differ too much to permit out-of-sample predictions of bailout expectations. To this extent, we want to make sure that we are for the larger part “interpolating” and not “extrapolating.” Table 5 therefore shows results for increasingly conservative subsamples of sound banks, for which we infer bailout expectations from distressed banks.

We first estimate the bailout Equation (1) for rolling windows that reflect the average election cycle in German states and allow for changing parameters to explain bailouts and annual predictions. Columns (3) and (4) show estimates of the distress Equation (2) for rolling windows of four and five years. Table A2 in the Appendix shows first-stage estimations for the five-year windows.²⁴ The significance of individual covariates differs across estimation windows, reflecting increasingly smaller subsample sizes. But the direction and magnitude of coefficients confirm the estimates for the entire sample in Table 4. This result holds especially for the identifying covariates. The diagnostic statistics confirm the validity of the identifying covariates for most rolling windows.

The five-year rolling-window approach implies that we obtain predicted bailouts only for the years starting in 1999 and thus lose some observations. For example, the first-stage bailout estimation uses lagged covariates to explain capital injections during the period of 1995–1999 to generate expectations pertaining to 1999. Bailout expectations are then generated only for those sound banks that are observed in 1999. We stack annual predictions from each rolling window to jointly estimate the distress Equation (2) for all bank-year observations.²⁵ For comparison, Table 5 shows all specifications for the identical period of 1999–2006. Both rolling-window approaches in Columns (3) and (4) confirm the existence of a significantly positive effect of bailout expectations on distress probabilities. The magnitude of the effect is lower, though. Remaining coefficients are likewise in line with the pooled sample estimates in Column (2). The average pseudo-Kleibergen-Paap

²⁴ To conserve space, all other first-stage estimations underlying Table 5 are provided in an Online Appendix.

²⁵ Note the assumption of constant parameters on $X_{i,t-1}$ and $M_{m,t-1}$ covariates to explain bank risk. Robustness checks that also specify the second-stage regression per rolling window yield qualitatively similar results, i.e., a significantly positive moral hazard coefficients, for most periods. These are available upon request.

F-statistic is somewhat weaker, but still above the rule-of-thumb critical value of ten.

One might interpret the rolling-window approach as banks “forgetting” about bailout decisions longer ago than the average election cycle, which may be inappropriate. Therefore, we also estimate the first-stage bailout regression based on a cumulating sample, which only considers preceding years to generate bailout expectations in any given year.²⁶ Column (5) in Table 5 shows the second-stage distress regression results. The magnitude of the moral hazard effect is between the pooled estimates in Column (2) and the shorter rolling windows in Columns (3) and (4). Other coefficient estimates are also barely affected in the distress equation.

To tackle the second main concern that sound and distressed banks might differ too much to permit inference on bailout expectations, Columns (6) through (8) show robustness checks regarding out-of-sample prediction. Throughout, we use rolling windows of five years and apply propensity score matching to determine a sample of sound banks that is sufficiently akin to that of distressed banks (Caliendo and Kopeinig 2008). Thereby, we ensure that inference of bailout expectations for the sound banks based on bailout regression estimates from distressed banks is warranted.

Specifically, propensity score matching uses a probit model to explain distress relative to being sound with all covariates specified in Equations (1) and (2), except with the identifying covariates. Based on resulting scores, we then determine the sound banks that are in the common support, i.e., those observations with a “treatment” probability conditional on the explanatory variables equal to that of being in the “nontreated,” sound group.²⁷ In Column (6), we determine the common support observations per rolling window (RW). The number of observations is reduced by only 271 observations, which indicates that most of the sound banks exhibit fairly similar characteristics compared to distressed banks. This result is in line with Koetter et al. (2007), who find only mild differences between the targets in voluntary and arranged mergers in German banking. Also, it confirms the insignificant differences of explanatory variables per event group shown in Table 3. Overall, predictions of bailout expectations to sound banks based on parameters estimated for distressed banks only seem defensible. The significantly positive bailout expectation corroborates this finding and confirms the existence of moral hazard for an alternative out-of-sample prediction of bailout expectations.

But banks might only infer expected bailout probabilities from those distressed peers that are supervised by the same regulator as themselves. Therefore, we further reduce the subsamples for which we determine the common support of sound banks and estimate the bailout regression to obtain

²⁶ Corresponding first-stage bailout regressions are shown in the Online Appendix.

²⁷ The balancing property is satisfied in all matchings shown in Table 5 at the 5% level.

predictions. Column (7) shows the results of the distress equation based on bailout expectations generated in each rolling window for each of the nine different Bundesbank jurisdictions (BBK) only.²⁸ For most jurisdictions and periods, the F-statistic is well above ten, and it equals sixty on average (Column (7), Table 5), clearly supporting the validity of instruments. The drawback of this approach is that much fewer observations in these increasingly fine subsamples reduce degrees of freedom. Given that supervisory guidelines are federal, we prefer the rolling-window approach without further subsampling per jurisdiction.

Sound observations are less often in the common support per jurisdiction and rolling window.²⁹ Thus, the number of observations to estimate the distress equation in Column (7) is substantially lower. The upshot of this yet stricter prediction rule is the robustness of the moral hazard effect. The coefficient of bailout expectations is 0.052 and thus very close to the effect found for the five-year rolling-window estimation in Column (3).

Since we prefer a linear probability model over a probit model to avoid identification based on functional form, another issue arises. A linear probability model is likely to predict probabilities above one or below zero. As a final robustness check regarding predicted bailout expectations, we show in Column (8) the results for the distress equation, excluding predicted bailout expectations that are above one or below zero. Only 729 observations fall outside this feasible range for the probability of a bailout, and exclusion of these observations does not alter the main result.

In sum, restricting the prediction of bailout expectations to banks in the common support and permitting the coefficients in the bailout equation to differ over the electoral cycle corroborates the main conclusion that bailout expectations exert an economically and statistically significant effect on risk. The point estimates for the moral hazard coefficient range between 0.035 and 0.218. Our preferred specification yields a coefficient of 0.071. Thus, an increase in the expected bailout probability by two standard deviations ($2 \times 20\% = 40\%$) causes an increase in the probability of distress of 2.8% ($40\% \times 0.071$). The most conservative estimate of 0.035 implies half the size of this effect, which is still economically significant, given an average distress probability of 8%.

4.2 Alternative risk measures

We argue that the probability of distress is a comprehensive and preferred measure for assessing multiple sources of risk managed by banks. But we

²⁸ A summary of the corresponding first-stage results and (common support) observations in the distress equation is provided in the Online Appendix. Each rolling window is a horizontal panel that is divided into nine columns corresponding to the respective Bundesbank branch in a jurisdiction. First, we show the number of observations in the estimated bailout regression, and then the corresponding pseudo-Kleibergen-Paap F-statistic.

²⁹ As shown in the last two lines per panel in Table A3 in the Online Appendix.

acknowledge that supervisory data on financial distress are rarely publicly available. To test whether and to what extent we can generalize the identified moral hazard effect, we show in Table 6 the relation between moral hazard and four alternative measures of bank risk (described in Table 3).

Because valid generation of bailout expectations is a major concern, we show in Columns (1)–(4) distress equation specifications using predictions for bailout expectations from the five-year rolling-window approach for banks that are in the common support only, corresponding to Column (6) in Table 5. Columns (5)–(8) show specifications that use the even stricter subsampling per rolling window and Bundesbank jurisdiction corresponding to Column (7) in Table 5 to obtain bailout expectations.

First, credit risk is an important individual risk driver for banks. A proxy for credit risk is the share of nonperforming loans (NPL; Demirgüç-Kunt and Detragiache 2002). We provide results using the NPL share as the dependent variable in Columns (1) and (5) of Table 6. The main result of increasing moral hazard due to higher bailout likelihood is confirmed for both alternatives to generate the bailout expectations shown.³⁰

Second, we specify the so-called z-scores, which Laeven and Levine (2009) define as $\frac{RoA + \frac{Equity}{RWA}}{\sigma_{RoA}}$. RoA denotes return on risk-weighted assets, RWA denotes risk-weighted assets, and σ_{RoA} is the standard deviation of RoA.³¹ Assuming that insolvency occurs when losses cannot be covered by equity, the probability of insolvency is $P(RoA < \frac{Equity}{RWA})$. If RoA follows a normal distribution, z-scores are inversely related to the probability of insolvency (Laeven and Levine 2009). Thus, z-scores reflect the number of standard deviations that a bank's RoA must fall below its expected value before equity is exhausted and the bank becomes insolvent. Lower z-scores therefore indicate riskier banks. Columns (2) and (6) in Table 6 confirm that higher bailout expectations increase risk, according to this measure.³²

Third, we gauge the sensitivity of banks to interest rate shocks with two measures available for a subsample of banks.³³ The first is the difference between the volume of assets with a fixed interest rate and liabilities with a fixed interest rate. Net fixed interest rate assets (NFIRA) are on average positive (€22 million). Larger NFIRA may insulate banks from interest rate

³⁰ Results are qualitatively identical for the alternative prediction methods discussed in Table 5.

³¹ σ_{RoA} is calculated on a rolling-window basis of the preceding three years, where possible. Using a constant standard deviation calculated for all available observations per bank does not affect the results. Here, total equity includes hidden reserves, which is why we do not specify the corresponding lag.

³² The smaller sample size is due to the limited availability of risk-weighted as opposed to gross total assets.

³³ Ideally, we would calculate duration gaps, but maturities of bank's assets and liabilities are not available.

Table 6
Alternative risk measures

Common support per:

Risk measure	Rolling window				Rolling window and BBK jurisdiction			
	(1) NPL	(2) Z-score	(3) NFIRA	(4) FIRG	(5) NPL	(6) Z-score	(7) NFIRA	(8) FIRG
Bailout expectation $\hat{\pi}$	1.663*** [0.351]	-2.86** [1.254]	1.541 [2.766]	0.154* [0.081]	2.79*** [0.819]	-0.135** [0.067]	0.036 [0.914]	-0.007 [0.041]
Total assets _{<i>t</i>-1}	-1.006*** [0.299]	-0.051 [0.300]	-0.75** [0.358]	-0.088*** [0.015]	0.062 [0.348]	-0.639* [0.341]	-0.377 [0.277]	-0.095*** [0.022]
Hidden reserves _{<i>t</i>-1}	2.381*** [0.726]		3.72*** [0.825]	0.066*** [0.024]	-0.784** [0.391]		3.655*** [0.518]	0.015 [0.022]
Nonperforming loan share _{<i>t</i>-1}		-0.063** [0.027]	-0.099* [0.058]	-0.000 [0.002]		-0.074*** [0.016]	-0.102** [0.045]	-0.000 [0.001]
Customer loan share _{<i>t</i>-1}	0.129*** [0.015]	0.009 [0.019]	-0.11*** [0.024]	0.005*** [0.001]	0.162*** [0.016]	-0.002 [0.024]	-0.122*** [0.022]	0.004** [0.002]
Return on equity _{<i>t</i>-1}	-0.028** [0.012]	0.104*** [0.027]	0.036* [0.021]	0.007*** [0.001]	-0.010 [0.018]	0.149*** [0.028]	0.031* [0.017]	0.007*** [0.001]
Cost-income ratio _{<i>t</i>-1}	0.015*** [0.006]	0.011 [0.019]	-0.074*** [0.010]	-0.003*** [0.000]	0.007 [0.006]	0.009 [0.021]	-0.068*** [0.008]	-0.003*** [0.001]
Liquid asset share _{<i>t</i>-1}	-0.099 [0.135]	-0.262* [0.135]	0.236 [0.220]	0.006 [0.010]	0.176*** [0.061]	-0.258** [0.128]	0.166 [0.266]	0.014 [0.010]
Regional market share _{<i>t</i>-1}	-0.061*** [0.012]	0.041*** [0.011]	-0.009 [0.024]	-0.000 [0.001]	-0.029** [0.012]	0.026 [0.016]	-0.009 [0.024]	-0.000 [0.001]
Corporate insolvency rate _{<i>t</i>-1}	-1.158* [0.637]	0.194 [0.360]	-0.355 [0.868]	-0.092*** [0.028]	0.279 [0.672]	0.354 [0.347]	0.394 [0.997]	-0.098*** [0.026]
Unemployment rate _{<i>t</i>-1}	-0.133*** [0.044]	-0.030 [0.050]	-0.027 [0.120]	-0.002 [0.004]	-0.040 [0.086]	0.037 [0.040]	-0.022 [0.125]	0.001 [0.005]
Real output per capita growth _{<i>t</i>-1}	-0.099*** [0.018]	0.041** [0.018]	0.055** [0.024]	-0.002 [0.001]	-0.033 [0.022]	0.021 [0.017]	0.08*** [0.017]	-0.002 [0.001]
Constant	-8.936*** [3.270]	15.414*** [1.606]	35.756*** [3.222]	1.322*** [0.158]	-3.704 [3.043]	11.951*** [1.532]	36.422*** [3.550]	1.421*** [0.171]
Observations	17,535	17,522	15,108	15,064	14,288	14,277	12,613	12,595
Centered R^2	0.178	0.137	0.183	0.238	0.151	0.147	0.168	0.213
Uncentered R^2	0.638	0.837	0.822	0.896	0.634	0.844	0.813	0.899

Second-stage risk regressions are with alternative dependent variables. Detailed definitions are given in the bottom panel of Table 2. NPL denotes the nonperforming loans share. The Z-score is defined as $(RoA + \text{Equity capital ratio})/\sigma_{RoA}$ (Laeven and Levine 2009). NFIRA denotes net fixed interest rate assets. FIRG fixed interest rate gap. Explanatory variables are defined in the top two panels of Table 2. Two-way clustered standard errors at the state and bank level (Petersen 2009) are in brackets. BBK jurisdictions refer to the nine Bundesbank regions. Fixed effects for states (“Bundesländer”), years, and banking sectors (commercial, savings, and cooperative). *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

shocks compared with banks with large shares of net variable interest assets. Higher expected bailout probabilities have no significant effect on NIRFA (Columns (3) and (7)), which may reflect the crudeness of this measure to capture interest rate shock sensitivity of banks.

Another measure of interest rate risk is the difference in average interest rates contracted for fixed assets and liabilities, respectively. The mean fixed interest rate gap (FIRG) for the entire sample is 1.3%. Banks earned a positive margin on their fixed income exposures. The coefficient of expected bailouts is significantly positive in Column (4), although only at the 10% level, and insignificant for the stricter expected bailout prediction rule shown in Column (8). In sum, for the smaller sample, in which risk sensitivity measures are available, we do not find evidence for moral hazard.

Overall, specific credit and insolvency risk measures confirm the existence of moral hazard effects due to bank bailouts.

4.3 Ownership

The three banking pillars differ regarding their ownership structure and the regional scope of banks' activities (Krahen and Schmidt 2004). Table 7 provides the subsample results to account for these differences.

Columns (1) to (3) show results for (regional) government-owned *savings banks*, which may pursue non-value-maximizing objectives (Sapienza 2004; Dinç 2005). Column (1) shows the effects for the subsample of local and large central savings banks (Landesbanken).³⁴ The coefficient of bailout expectations is insignificant. Moral hazard thus is not a phenomenon driven per se by government ownership in German banking.

But Columns (2) and (3) show that the effect differs for regional savings banks and (inter)nationally operating Landesbanken. The absence of a significant moral hazard effect for small, regional savings resembles the results of Barth et al. (2004). They find no effect of government ownership on bank risk after controlling for regulatory traits and supervisory practices. In contrast, we find a large and significantly positive effect of bailout expectations on risk taking for the large Landesbanken. Whereas the small sample size and weak significance call for care in drawing strong inference, the existence of moral hazard is in line with the failure of numerous Landesbanken due to the 2007–2008 financial crisis (see also Puri et al. 2011).³⁵

Columns (4) to (7) show subsample results for commercial and cooperative banks, which are privately owned. Bailout expectations significantly increase risk for this group of banks as a whole (Column (4)). Private ownership

³⁴ BayernLB, Bremer Landesbank, HSH Nordbank, Landesbank Baden-Württemberg, Landesbank Berlin, Landesbank Hessen-Thüringen, Norddeutsche Landesbank, Landesbank Saar, and WestLB.

³⁵ WestLB, SachsenLB, BayernLB, and HSH Nordbank were rescued by the federal government.

Table 7
Moral hazard and bank ownership

	Government-owned savings banks			Private commercial and cooperative banks			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	All	Regional	Large	All	Regional	Commercial	Coops
Bailout expectations $\hat{\pi}$	0.017 [0.029]	0.016 [0.030]	0.548* [0.319]	0.106*** [0.025]	0.101*** [0.025]	0.057 [0.060]	0.089** [0.035]
Total assets _{<i>t</i>−1}	−0.005 [0.003]	−0.003 [0.003]		−0.003 [0.007]	0.000 [0.007]	−0.012* [0.007]	0.019** [0.010]
Hidden reserves _{<i>t</i>−1}	−0.026*** [0.010]	−0.028*** [0.010]	−0.692* [0.416]	−0.062*** [0.014]	−0.063*** [0.014]	−0.012 [0.021]	−0.057*** [0.015]
Nonperforming loan share _{<i>t</i>−1}	0.001 [0.001]	0.001* [0.001]	0.001 [0.003]	0.002*** [0.001]	0.002*** [0.001]	−0.001** [0.000]	0.002*** [0.001]
Customer loan share _{<i>t</i>−1}	0.002** [0.001]	0.001** [0.001]	−0.005 [0.003]	0.001** [0.000]	0.001** [0.000]	0.000 [0.000]	0.002*** [0.001]
Return on equity _{<i>t</i>−1}	−0.004*** [0.001]	−0.004*** [0.001]	−0.005 [0.004]	−0.004*** [0.001]	−0.004*** [0.001]	−0.002*** [0.000]	−0.006*** [0.001]
Cost-income ratio _{<i>t</i>−1}	0.002 [0.001]	0.003* [0.002]	0.002 [0.003]	0.001* [0.000]	0.001* [0.000]	0.001 [0.000]	0.001*** [0.000]
Liquid asset share _{<i>t</i>−1}	0.008** [0.004]	0.005* [0.003]	0.09* [0.052]	0.004 [0.003]	0.004 [0.003]	−0.002 [0.002]	0.009** [0.005]
Regional market share _{<i>t</i>−1}	0.000 [0.000]	0.000 [0.000]	−0.002 [0.000]	0.000 [0.002]	0.001 [0.001]	0.000 [0.001]	0.000 [0.000]
Corporate insolvency rate _{<i>t</i>−1}	0.014 [0.016]	0.018 [0.016]	−0.228** [0.091]	0.062*** [0.018]	0.062*** [0.018]	0.027 [0.038]	0.053*** [0.020]
Unemployment rate _{<i>t</i>−1}	0.002 [0.003]	0.002 [0.003]	0.031 [0.023]	0.005* [0.003]	0.005* [0.003]	0.007 [0.005]	0.005* [0.003]
Real output per capita growth _{<i>t</i>−1}	0.000 [0.001]	0.000 [0.001]	−0.004 [0.005]	−0.002* [0.001]	−0.002* [0.001]	0.002 [0.002]	−0.002* [0.001]
Constant	−0.069 [0.116]	−0.121 [0.125]	−0.649 [0.693]	0.138 [0.088]	0.145 [0.089]	0.56*** [0.127]	0.065 [0.097]
Observations	3,963	3,869	94	13,572	13,527	1,382	12,190
Centered <i>R</i> ²	0.109	0.094	0.737	0.112	0.112	0.102	0.126
Uncentered <i>R</i> ²	0.137	0.121	0.751	0.208	0.209	0.169	0.224

Linear probability model with an indicator variable is equal to one if a bank was distressed for subsamples of banks with different ownership and incorporation. Explanatory variables are defined in Table 2. Bailout expectations are based on the five-year rolling window for the first-stage regressions with out-of-sample predictions only to banks in the common support. Government-owned savings banks (“Sparkassen”) are divided into regional and (inter)nationally active, large savings banks (“Landesbanken”). Commercial and cooperative banks are privately owned and divided into smaller banks that operate de facto on regional markets only, commercial banks only, and mutually owned cooperatives. Explanatory variables are defined in the top two panels of Table 3. Two-way clustered standard errors at the state and bank level are in brackets (Petersen 2009). Fixed effects are for states (“Bundesländer”), years, and banking sectors (commercial, savings, and cooperative), where appropriate. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

therefore appears not to be a panacea to avoid excessive risk taking due to the existence of safety nets. Column (5) compares results from a subsample that excludes the largest banks from the commercial and cooperative pillars. These banks confine their operations *de facto* to regionally demarcated markets. They are all legally independent institutions, for which group holdings highlighted by Dinç (2006) to be important for the governance of Japanese banks are absent. The estimated moral hazard effect remains significantly positive. Thus, moral hazard is not only a problem if banks are too-big-to-fail, but might also be the result of too-many-to-fail, as argued by Acharya and Yorulmazer (2007) and Brown and Dinç (2011).

Columns (6) and (7) show that the large number of small cooperative banks, which are mutually owned by their members, drives most of the moral hazard effect due to safety nets. Large shareholdings of managers increase risk taking for large and listed U.S. bank holding companies (BHC; Gorton and Rosen 1995). Members of cooperatives are the depositors, and they are dispersed. Hence, the importance of (dominant) ownership by managers among U.S. BHCs is unlikely to play an important role in the risk taking of German cooperative banks.

Among large, privately owned banks, primarily the so-called big four in the commercial sector and one central cooperative bank are (inter)nationally active.³⁶ This subsample is too small to permit estimation. To assess the potentially important role of stricter publication requirements and capital markets for risk taking, we split the sample according to public and private incorporation. The moral hazard effect is significant for both groups.³⁷

In sum, the moral hazard effect is driven by the many mutually owned cooperatives. Government ownership of large Landesbanken seems problematic, but we find no evidence of moral hazard for regional savings banks.

5. Can Supervisors Mitigate Moral Hazard?

The BaFin can intervene with different degrees of severity in the going concerns of financial institutions that it deems at risk. This scenario begs a question: Can supervisory actions mitigate moral hazard? We obtain data from the supervisory department of the Bundesbank, which collects all intervention measures, and we categorize the data into four groups. Interventions differ from capital preservation measures, which are conducted by the insurance schemes of each banking sector in consultation with the auditors of the respective banking organizations (e.g., DSGV or BVR). Interventions, instead, are actions taken solely by the official authority (BaFin) in pursuit of its mandate to ensure financial stability.

³⁶ Deutsche Bank, Commerzbank, HypoVereinsbank, Postbank, and DZ Bank as of 2010.

³⁷ Not reported; results are available on request.

Table A3 depicts the frequency and detailed composition per intervention category. Descriptive statistics show that sound banks are rather frequently subject to interventions based on their behavior, too. For example, a bank that has gambled and won might receive a warning but not require a capital injection. Thus, interventions neither mimic capital preservation measures taken by insurance schemes nor are an ultimate resort of the regulator in terms of forcing banks to exit through mergers.

Table 8 shows the results of the distress Equation (2) augmented with interaction terms between intervention indicators and bailout expectations.³⁸

Across all specifications, the parameter for bailout expectations is significant, confirming the main result that moral hazard prevails. We focus here on the interaction effects between intervention categories and bailout expectations to test if moral hazard is mitigated by supervisory actions.

Supervisory actions labeled Warnings in Column (1) of Table 8 concern hearings, official disapproval, or threats of further measures. These fairly weak interventions account for the vast majority of prudential supervisory actions (see Table A3). These weak measures have the undesirable effect of amplifying moral hazard, which outweighs the direct benefits from officially cautioning a bank. Potentially, prudential warnings signal to the bank that regulators consider the bank sufficiently important to warrant additional monitoring efforts and perhaps even rescue the bank if it becomes distressed. To some extent, the failure of warnings to effectively reduce risk taking is consistent with Hoshi and Kashyap (2010), who argue that successful recovery of distressed banks requires sturdy regulatory action.

The second category, Management, comprises such sturdier interventions that interfere directly with staffing decisions of the bank, such as the rejection of branch director appointments, limitations of the permitted scope of managers' responsibilities, or appointment of a supervisory officer. These interventions occur less often. Whereas the interaction term shown in Column (2) of Table 8 has a negative sign, it is not statistically different from zero.

Alternative strong measures restrict or prohibit certain activities of the bank and therefore primarily affect owners, debtholders, and potential borrowers instead of the management. This third category includes restrictions and prohibitions of dividend distribution, new lending, or deposit taking, as well as orders to restructure processes. Table A3 shows that these events occur frequently and often apply to banks that eventually exit. However, the negative interaction term in Column (3) of Table 8 is also insignificant.

³⁸ Bailout expectations are generated using the five-year rolling window and are generated for sound banks that are on the common support. All covariates and fixed effects are specified but not reported to conserve space.

Table 8
Regulatory intervention and moral hazard

	Warnings (1)	Management (2)	Restrictions (3)	Penalties (4)	Count (5)	All (6)	NPL (7)	z-score (8)
Bailout expectations $\hat{\pi}$	0.064*** [0.019]	0.071*** [0.019]	0.073*** [0.019]	0.071*** [0.019]	0.058*** [0.020]	0.058*** [0.020]	1.663*** [0.353]	-2.727** [1.236]
Warnings $\times \hat{\pi}$	0.499*** [0.092]					0.080 [0.090]	3.938 [2.518]	-0.729 [2.739]
Management $\times \hat{\pi}$		-0.233 [0.426]				-0.667** [0.279]	-4.643 [9.683]	15.153 [10.746]
Restrictions $\times \hat{\pi}$			-0.412 [0.311]			-0.589** [0.298]	-16.986*** [5.213]	6.306 [13.451]
Penalties $\times \hat{\pi}$				0.464*** [0.157]		0.312* [0.182]	2.447 [5.206]	-50.694*** [8.874]
Count $\times \hat{\pi}$					0.22*** [0.046]	0.219*** [0.041]	1.9*** [0.647]	-0.741 [1.441]
Warnings	-0.276*** [0.060]					-0.129** [0.061]	-1.614 [2.021]	-0.648 [2.347]
Management		0.268 [0.275]				0.442** [0.211]	0.809 [7.444]	-14.356* [7.836]
Restrictions			0.72*** [0.242]			0.722*** [0.233]	12.951*** [2.856]	-7.574 [10.583]
Penalties				-0.676*** [0.156]		-0.653*** [0.172]	-2.210 [4.852]	47.110*** [8.011]
Count					-0.082** [0.037]	-0.069** [0.032]	-1.444*** [0.463]	-1.383 [1.150]
Observations	17,535	17,535	17,535	17,535	17,535	17,535	17,534	17,522
Centered R^2	0.110	0.107	0.110	0.107	0.121	0.125	0.179	0.140
Uncentered R^2	0.191	0.188	0.191	0.188	0.201	0.204	0.638	0.838

Linear probability model for second-stage risk regressions specifies prudential intervention indicators and their interaction with moral hazard. The dependent variable in Columns (1) through (6) is an indicator variable equal to one if a bank was distressed. The dependent variable in Column (7) is the nonperforming loan share and measures credit risk. The dependent variable in Column (8) is the z-score (Laeven and Levine 2009) and measures insolvency risk. Explanatory variables and alternative risk measures are defined in Table 2. Direct and interaction terms refer to four different types of intervention. “Warnings” comprise events, such as hearings and official letters from the supervisor. “Management” comprises measures targeting bank managers, such as replacements. “Restrictions” comprise interventions prohibiting certain business activities, such as granting loans or taking deposits. “Penalties” are fines to managers or institutions. The detailed intervention group composition is shown in Table A3. The intervention count is the sum of intra-annual interventions per bank. Other explanatory variables are specified, but not reported, and defined in the two top panels of Table 2. Two-way clustered standard errors at the state and bank level are in brackets (Petersen 2009). Fixed effects are for states (“Bundesländer”), years, and banking sectors (commercial, savings, and cooperative). *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

The fourth category, Penalties, contains measures that require the bank or managers to pay fines. These interventions are rare and pertain mostly to sound banks. Interaction effects shown in Column (4) mimic those of warnings; levying fines increases additional risk taking due to moral hazard.

Importantly, some banks receive more than one intervention, potentially of different types, during a given year. Column (5) shows results for a count variable per bank and year of received interventions. More interventions also increase moral hazard. It may reflect that heightened regulatory attention, expressed by multiple actions, signals to the bank that it is considered sufficiently important to potentially be bailed out when in distress.

Another implication of multiple interventions is that regulators seem to use different actions simultaneously. The joint specification of all four categories and the count indicator is shown in Column (6). The effects of both sturdy types of actions, against management directly as well as restrictions to the bank, are now significantly negative. Increasing moral hazard due to warnings and penalties is not, or is only barely, significant. This result suggests that committed prudential regulatory intervention can mitigate the moral hazard arising from the existence of safety nets. Related, numerous interventions amplify moral hazard, confirming concerns about piecemeal intervention policies (Hoshi and Kashyap 2010).

Column (7) shows that the mitigating effect of managerial interventions on nonperforming loans is insignificant. But the interaction between bailout expectations and restrictions remains significantly negative, and a larger number of supervisory interventions increases moral hazard. For the z-score used in Column (8) as the measure of risk, we do not find significant relations on moral hazard mitigation.

In summary, for our preferred distress indicator, moral hazard due to safety nets can be reduced by the sturdier interventions targeting the bank's management or imposing restrictions. Weak interventions are ineffective, and many supervisory actions are even counterproductive.

6. Conclusion

We test if bailout expectations increase moral hazard in the banking industry in terms of excessive risk-taking behavior. To this end, we develop a simultaneous structural equations model. In a first stage, we estimate expected bailout probabilities, and in a second stage we regress measures for risk-taking behavior on the estimated bailout expectations. Moral hazard is estimated as the sensitivity of distress probabilities with respect to an increase in the expected bailout probability.

To separate bad luck from bad behavior, we use regional political factors as identifying covariates that explain bailouts, but do not directly affect

risk-taking behavior. We combine these identifying covariates with a unique sample provided by the German central bank that contains detailed information on capital injections, regulatory interventions, and distressed exits at the bank level. The sample includes 3,554 German banks from 1995 until 2006, covering virtually the entire population. The share of distressed banks is 8%, of which 75% are bailed out. For our preferred approach in estimating bailout expectations, an increase of bailout expectations by two standard deviations implies an increase in predicted distress probabilities by 2.8%. Given the mean distress probability of 8%, this effect is substantial.

The result—safety nets in banking fuel moral hazard—is robust across a range of methods to predict bailout expectations for sound banks. Alternative measures of risk taking corroborate the significance of moral hazard for credit and insolvency risk. Estimations for different banking groups show that the moral hazard effect is not driven by government-owned, regional savings banks. Instead, the results are predominantly due to the largest group of mutually owned cooperative banks. However, we find weak evidence of a moral hazard effect for the group of large government-owned central savings banks, so-called Landesbanken.

Selected supervisory interventions can mitigate moral hazard, which has potentially important policy implications. Moral hazard is significantly reduced if interventions aim directly at the bank's management or involve outright restrictions of its business. Sturdy interventions are effective. But weaker measures, such as warnings that are more often used, are ineffective in reducing moral hazard. In fact, multiple interventions at a bank may even foster moral hazard. A potential reason is that such a piecemeal approach by supervisors may signal to the bank that it is considered important enough to be saved in case of distress.

Appendix

Table A
Spatial and temporal distribution of identifying covariates

State/Year	1995	1996	1997	1998	1999	2000	2001	2002	2003	2004	2005	2006
State election indicator												
Baden-Württemberg	0	1	0	0	0	0	1	0	0	0	0	1
Bavaria	0	0	0	1	0	0	0	0	1	0	0	0
Berlin	1	0	0	0	1	0	1	0	0	0	0	1
Bremen	0	0	0	0	1	0	0	0	0	1	0	0
Hamburg	1	0	0	0	1	0	0	0	1	0	0	0
Hessia	0	0	1	0	0	0	1	0	0	1	0	0
North Rhine Westphalia	1	0	0	0	1	0	0	0	1	0	0	0
Lower Saxony	1	0	0	0	0	1	0	0	0	0	1	0
Rhineland-Palatinate	0	0	0	1	0	0	0	1	0	0	0	1
Saarland	0	0	0	1	0	0	0	0	1	0	0	0
Schleswig-Holstein	0	1	0	0	0	0	1	0	0	0	0	1
Mecklenburg-Western Pomerania	0	0	0	0	1	0	0	0	0	1	0	0
Brandenburg	0	0	0	1	0	0	0	1	0	0	0	1
Saxony-Anhalt	0	0	0	0	1	0	0	0	0	1	0	0
Thuringia	0	1	0	0	0	1	0	0	0	0	1	0
Saxony	0	0	0	0	1	0	0	0	0	1	0	0
Vote share difference between cabinet party/coalition and most powerful opposition party												
Baden-Württemberg	60.5	28.1	28.1	27.9	27.9	27.5	21.9	21.7	21.4	21.4	21.4	31.1
Bavaria	23.8	23.8	23.8	27.4	27.4	27.4	27.3	27.0	44.0	44.2	44.3	44.1
Berlin	46.4	46.4	46.4	46.4	45.4	45.4	15.0	28.5	28.5	28.5	28.5	22.9
Bremen	34.1	34.1	34.0	33.7	42.3	42.2	42.2	42.0	42.4	22.5	22.3	22.2
Hamburg	54.0	54.0	54.0	53.7	70.3	70.3	70.3	70.3	59.4	59.4	59.4	59.4
Hessia	15.3	15.3	19.5	19.5	19.5	19.5	13.4	13.4	13.4	16.7	16.7	16.7
North Rhine Westphalia	10.0	10.2	9.9	10.0	9.3	9.5	9.4	9.5	18.7	18.8	18.8	18.8
Lower Saxony	11.4	11.4	11.4	11.5	11.6	7.0	7.2	7.5	7.4	7.4	19.3	19.1
Rhineland-Palatinate	45.6	45.1	44.9	26.2	27.1	26.9	27.1	25.3	25.8	25.8	25.8	41.5
Saarland	6.0	6.1	6.1	7.7	7.7	8.1	7.7	8.3	27.3	26.9	27.3	27.2
Schleswig-Holstein	11.7	8.8	8.9	8.7	8.3	8.7	16.6	16.8	17.3	17.6	17.5	13.0
Mecklenburg-Western Pomerania	10.7	10.7	10.6	10.4	1.6	1.7	1.3	1.2	1.4	17.0	17.2	17.2
Brandenburg	3.8	4.0	4.3	13.5	13.5	13.4	13.6	30.4	30.4	30.4	30.3	33.7
Saxony-Anhalt	41.9	42.1	42.3	42.7	47.0	46.9	47.0	46.8	46.9	27.5	27.7	27.9
Thuringia	11.2	9.3	9.4	9.6	9.7	13.5	13.7	13.8	13.7	13.8	71.5	71.6
Saxony	57.5	57.3	57.0	56.8	33.2	32.9	32.9	32.8	32.7	29.0	29.0	29.2
Federal and state prime minister from the same party												
Baden-Württemberg	1	1	1	0	0	0	0	0	0	0	1	1
Bavaria	1	1	1	0	0	0	0	0	0	0	1	1
Berlin	1	1	1	0	0	0	1	1	1	1	0	0
Bremen	0	0	0	1	1	1	1	1	1	1	0	0
Hamburg	0	0	0	1	1	1	1	1	1	1	0	0
Hessia	0	0	0	1	1	1	0	0	0	0	1	1
North Rhine Westphalia	0	0	0	1	0	0	0	0	0	0	1	1
Lower Saxony	0	0	0	1	1	1	1	1	1	1	1	1
Rhineland-Palatinate	1	1	1	1	1	1	1	1	1	1	0	0
Saarland	0	0	0	1	1	1	1	1	0	0	1	1
Schleswig-Holstein	0	0	0	1	1	1	1	1	1	1	0	0
Mecklenburg-Western Pomerania	0	0	0	1	0	0	0	0	0	0	1	1
Brandenburg	0	0	0	1	1	1	1	0	0	0	1	1
Saxony-Anhalt	1	1	1	0	0	0	0	0	0	0	1	1
Thuringia	0	0	0	1	1	1	1	1	1	1	1	1
Saxony	1	1	1	0	0	0	0	0	0	0	1	1

This table details data on the identifying covariates over time and per state. The first panel indicates elections with a one. The second panel shows the mean vote share differences across municipalities per state and year. The third panel indicates if the state and federal prime minister are from the same party with a one.

Table A2
First-stage bailout regression: Five-year rolling window

Sample from up and until	5-year rolling window							
	1995 1999	1996 2000	1997 2001	1998 2002	1999 2003	2000 2004	2001 2005	2002 2006
Bank-specific and regional macro								
Total assets _{<i>t</i>-1}	0.041 [0.032]	0.048 [0.034]	0.069 [0.044]	0.119*** [0.037]	0.09** [0.036]	0.064** [0.026]	0.073*** [0.027]	0.067*** [0.020]
Hidden reserves _{<i>t</i>-1}	-0.088 [0.070]	-0.145 [0.093]	-0.246*** [0.093]	-0.328*** [0.066]	-0.273*** [0.028]	-0.289*** [0.025]	-0.155*** [0.040]	-0.052 [0.051]
Nonperforming loan share _{<i>t</i>-1}	0.003*** [0.001]	0.003*** [0.001]	0.003 [0.001]	0.004*** [0.001]	0.003 [0.002]	0.002 [0.001]	0.004*** [0.001]	0.003*** [0.001]
Customer loan share _{<i>t</i>-1}	0.001 [0.001]	0.002 [0.001]	0.001 [0.001]	0.001 [0.001]	0.002 [0.001]	0.002* [0.001]	0.001 [0.001]	0.001 [0.001]
Return on equity _{<i>t</i>-1}	0.001 [0.001]	0.001 [0.001]	0.000 [0.002]	0.001 [0.001]	0.000 [0.001]	0.001 [0.001]	0.001 [0.001]	0.002* [0.001]
Cost-income ratio _{<i>t</i>-1}	0.000 [0.001]	-0.001 [0.001]	-0.001*** [0.000]	-0.001*** [0.000]	-0.001*** [0.000]	-0.001*** [0.000]	-0.001*** [0.000]	-0.001** [0.000]
Liquid asset share _{<i>t</i>-1}	0.005 [0.010]	0.011 [0.012]	0.020** [0.010]	0.011 [0.017]	0.017 [0.015]	0.031** [0.014]	0.031* [0.016]	0.023 [0.019]
Regional market share _{<i>t</i>-1}	0.002** [0.001]	0.003** [0.001]	0.004*** [0.001]	0.003** [0.001]	0.002 [0.001]	0.003** [0.001]	0.002* [0.001]	0.002* [0.001]
Corporate insolvency rate _{<i>t</i>-1}	0.103 [0.089]	0.023 [0.073]	-0.077 [0.107]	0.082 [0.118]	0.016 [0.100]	0.041 [0.054]	0.09** [0.039]	0.089** [0.042]
Unemployment rate _{<i>t</i>-1}	0.002 [0.005]	0.006 [0.005]	0.008 [0.007]	0.008 [0.007]	0.012 [0.007]	0.012*** [0.005]	0.009** [0.004]	0.006* [0.004]
Real output per capita growth _{<i>t</i>-1}	-0.002 [0.003]	0.008* [0.004]	0.008** [0.003]	0.005* [0.003]	0.006** [0.003]	0.004 [0.003]	0.004 [0.003]	0.006* [0.003]
Constant	0.584*** [0.185]	0.608*** [0.128]	0.788*** [0.202]	0.475** [0.206]	0.523** [0.227]	0.409*** [0.128]	0.354*** [0.087]	0.478*** [0.133]

(Continued)

Table A2
(Continued)

Identifying variables								
Vote share difference _{<i>t</i>}	−0.001** [0.001]	−0.002 [0.002]	−0.001* [0.001]	−0.002** [0.001]	−0.002* [0.001]	−0.002** [0.001]	−0.001** [0.001]	−0.001** [0.001]
Election _{<i>t</i>}	−0.025* [0.013]	−0.035** [0.015]	−0.019 [0.025]	−0.039*** [0.014]	−0.041** [0.018]	−0.070*** [0.027]	−0.060*** [0.021]	−0.072*** [0.026]
Same party _{<i>t</i>}	−0.053* [0.029]	−0.065** [0.027]	−0.058** [0.026]	−0.089 [0.068]	−0.123 [0.105]	−0.126** [0.054]	−0.015 [0.031]	−0.01 [0.033]
Observations	1,129	1,162	1,208	1,200	1,171	1,096	1,006	869
Centered <i>R</i> ²	0.112	0.111	0.111	0.13	0.134	0.138	0.139	0.142
Pseudo-Kleibergen-Paap F-statistic	11.828	11.29	8.068	10.09	11.33	11.76	13.88	16.79
Pseudo-Hansen J-statistic	2.082	0.122	2.515	1.02	1.328	2.523	3.086	3.134
<i>p</i> -value of Hansen	0.353	0.941	0.284	0.601	0.515	0.283	0.214	0.209

Linear probability model with an indicator variable equal to one if a bank was bailed out and zero if it exited. Estimation sample is based on a rolling window of five years. Explanatory variables are defined in Table 2. Vote share differences are the share of votes obtained by the majority party/coalition per municipality during state parliament elections less the vote share of the strongest opposition party in the municipality. Elections are an indicator variable equal to one if state parliament elections were held in the municipality in which the bank is headquartered during that year. Same party is an indicator variable equal to one if the party of the prime minister of the state in which a bank is headquartered is equal to the party of the ruling federal prime minister. The F-test statistic is the pseudo-Kleibergen-Paap test for weak underidentification pertaining to the excluded covariates in the respective columns (Baum et al. 2007). The J-statistic is the pseudo-Hansen test statistic for over identification (Baum et al. 2007). Two-way clustered standard errors at the state and bank levels are in brackets (Petersen 2009). Fixed effects are for states (“Bundesländer”), years, and banking sectors (commercial, savings, and cooperative). *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

Table A3
Detailed components of regulatory intervention categories

Measure	Sound	Bailout	Exit	Total
Warnings	708	30	193	931
Hearing about banking charter revocation	0	2	0	2
Threat of measures, according to §45 KWG	6	1	8	15
Disapproval hearing	24	0	11	35
Threat of measures, according to §36 KWG	34	3	9	46
Hearing dismissal request	15	7	39	61
Disapproval	69	6	21	96
Admonishment hearing	134	6	31	171
Warning branch director	193	4	32	229
Serious letter	233	1	42	276
Management	22	11	5	38
Limitation of senior management competencies	1	1	1	1
Disapproval branch director	5	0	0	7
Supervisor appointed, according to §46 KWG	2	4	1	7
Dismissal branch director	14	6	3	23
Restrictions	40	42	175	257
Restricting deposit taking	0	1	0	1
Measures, according to §6(3) KWG	1	0	0	1
Prohibiting profit distribution	0	1	1	2
Restricting new lending	3	1	3	7
Correction position, according to §10(3b) KWG	2	5	5	12
Other measures, according to §46 KWG	4	11	11	26
Prohibiting new lending	18	5	16	39
Restructuring	12	18	139	169
Penalties	7	3	0	10
Fine branch director	4	0	0	4
Fine financial institution	3	3	0	6
Total	777	86	373	1,236

The table shows the number of measures categorized into the four intervention groups. Multiple events are possible and accounted for in the regressions by a count of intervention variables per bank and year. KWG refers to the German Banking Act (Kreditwesengesetz). Provisions, according to the respective sections of the Banking Act, are categorized as shown, depending on the specific nature of the regulatory measure taken.

References

- Acharya, V. V., and T. Yorulmazer. 2007. Too Many to Fail—An Analysis of Time Inconsistency in Bank Closure Policies. *Journal of Financial Intermediation* 16:1–31.
- Anselin, L. 1988. *Spatial Econometrics: Methods and Models*. Dordrecht, The Netherlands: Kluwer Academic Publishers.
- Barth, J., G. Caprio, and R. Levine. 2004. Bank Regulation and Supervision: What Works Best? *Journal of Financial Intermediation* 13:205–48.
- Baum, C. F., M. E. Schaffer, and S. Stillmann. 2007. Enhanced Routines for Instrumental Variables/Generalized Method of Moments Estimation and Testing. *Stata Journal* 7:465–506.
- Boyd, J. H., C. Chang, and B. D. Smith. 2004. Deposit Insurance and Bank Regulation in a Monetary Economy: A General Equilibrium Exposition. *Economic Theory* 24:741–67.
- Brown, C. O., and I. S. Dinc. 2005. The Politics of Bank Failures: Evidence from Emerging Markets. *Quarterly Journal of Economics* 120:1413–44.
- . 2011. Too Many to Fail? Evidence of Regulatory Forbearance When the Banking Sector Is Weak. *Review of Financial Studies* 24:1378–405.

- Caliendo, M., and S. Kopeinig. 2008. Some Practical Guidance for the Implementation of Propensity Score Matching. *Journal of Economic Surveys* 22:31–72.
- Cordella, T., and E. L. Yeyati. 2003. Bank Bailouts: Moral Hazard vs. Value Effect. *Journal of Financial Intermediation* 12:300–330.
- Cornelli, F., Z. Kominek, and A. Ljungqvist. 2011. Monitoring Managers: Does It Matter? 6th FIRS Conference, mimeo.
- Demirgüç-Kunt, A., and E. Detragiache. 2002. Does Deposit Insurance Increase Banking System Stability? An Empirical Investigation. *Journal of Monetary Economics* 49:1373–406.
- Deutsche Bundesbank. 2007. Financial Stability Report 2007. Frankfurt a.M.: Deutsche Bundesbank.
- Diamond, D. W., and P. H. Dybvig. 1983. Bank Runs, Deposit Insurance, and Liquidity. *Journal of Political Economy* 91:401–19.
- Dinç, I. S. 2005. Politicians and Banks: Political Influences on Government-owned Banks in Emerging Markets. *Journal of Financial Economics* 77:453–79.
- . 2006. Monitoring the Monitors: The Corporate Governance in Japanese Banks and Their Real Estate Lending in the 1980s. *Journal of Business* 79:3057–81.
- Dinç, I. S., and N. Gupta. 2011. The Decision to Privatize: Finance and Politics. *Journal of Finance* 66:241–69.
- Freixas, X., and J.-C. Rochet. 2011. Taming SIFIS. Working Paper, Zurich University.
- Gale, D., and X. Vives. 2002. Dollarization, Bailouts, and the Stability of the Banking System. *Quarterly Journal of Economics* 117:467–502.
- Gorton, G., and R. Rosen. 1995. Corporate Control, Portfolio Choice, and the Decline in Banking. *Journal of Finance* 50:1377–420.
- Gorton, G., and L. Huang. 2004. Liquidity, Efficiency, and Bank Bailouts. *American Economic Review* 94: 455–83.
- Gropp, R., and J. Vesala. 2004. Deposit Insurance, Moral Hazard, and Market Monitoring. *Review of Finance* 8:571–602.
- Gropp, R., H. Hakenes, and I. Schnabel. 2011. Competition, Risk Shifting, and Public Bailout Policies. *Review of Financial Studies* 24:2084–120.
- Grossman, R. 1992. Deposit Insurance, Regulation, and Moral Hazard in the Thrift Industry: Evidence from the 1930s. *American Economic Review* 82:800–821.
- Hazlett, D. 1997. Deposit Insurance and Regulation in a Diamond-Dybvig Banking Model with a Risky Technology. *Economic Theory* 9:453–70.
- Hoshi, T., and A. K. Kashyap. 2010. Will the U.S. Bank Recapitalization Succeed? Eight Lessons from Japan. *Journal of Financial Economics* 97:398–417.
- Huang, R. 2008. The Real Effect of Bank Branching Deregulation: Comparing Contiguous Counties across U.S. State Borders. *Journal of Financial Economics* 87:678–705.
- Kareken, J. H., and N. Wallace. 1978. Deposit Insurance and Bank Regulation: A Partial-equilibrium Exposition. *Journal of Business* 51:413–38.
- Karels, G. V., and C. A. McClatchey. 1999. Deposit Insurance and Risk-taking Behavior in the Credit Union Industry. *Journal of Banking and Finance* 23:105–34.
- Keeley, M. C. 1990. Deposit Insurance, Risk, and Market Power in Banking. *American Economic Review* 80:1183–200.
- Khwaja, A. I., and A. Mian. 2005. Do Lenders Favor Politically Connected Firms? Rent Provision in an Emerging Financial Market. *Quarterly Journal of Economics* 120:1378–411.

- Koetter, M., J. W. B. Bos, F. Heid, J. W. Kolari, C. J. M. Kool, and D. Porath. 2007. Accounting for Distress When Predicting Bank Mergers. *Journal of Banking and Finance* 32:3200–17.
- Krahnhen, J. P., and R. H. Schmidt, eds. 2004. *The German Financial System*. Oxford: Oxford University Press.
- Laeven, L., and R. Levine. 2009. Bank Governance, Regulation, and Risk Taking. *Journal of Financial Economics* 93:259–75.
- Murphy, K. M., and R. H. Topel. 2002. Estimation and Inference in Two-step Econometric Models. *Journal of Business and Economic Statistics* 20:88–97.
- O'Hara, M., and S. Wayne. 1990. Deposit Insurance and Wealth Effects: The Value of "Too Big to Fail." *Journal of Finance* 45:1587–600.
- Petersen, M. 2009. Estimating Standard Errors in Finance Panel Data Sets: Comparing Approaches. *Review of Financial Studies* 22:435–80.
- Puri, M., J. Rocholl, and S. Steffen. 2011. Global Retail Lending in the Aftermath of the U.S. Financial Crisis: Distinguishing Between Supply and Demand Effects. *Journal of Financial Economics* 100:556–78.
- Repullo, R. 2000. Who Should Act as Lender of Last Resort? An Incomplete Contracts Model. *Journal of Money, Credit, and Banking* 32:580–605.
- Repullo, R., and D. Martinez-Miera. 2010. Does Bank Competition Reduce the Risk of Bank Failure? *Review of Financial Studies* 23:3638–64.
- Sapienza, P. 2004. The Effects of Government Ownership on Bank Lending. *Journal of Financial Economics* 72:357–84.
- Stock, J. H., J. H. Wright, and M. Yogo. 2002. A Survey of Weak Instruments and Weak Identification in Generalized Methods of Moments. *Journal of Business and Economic Statistics* 20:518–29.
- Strahan, P. E., and T. Rice. 2010. Does Credit Competition Affect Small-firm Finance? *Journal of Finance* 65:861–89.
- Wheelock, D. C., and P. W. Wilson. 1995. Explaining Bank Failures: Deposit Insurance, Regulation, and Efficiency. *Review of Economics and Statistics* 77:689–700.